Capacity and Composition of Shared Transposition Decoders

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Abstract

A shared transposition decoder sees a cyclic column rotation of the complete matrix and its receiver class. We study the dimension of linear sources on which such a rule is correct and the composition of complete rules. For two classes, arbitrary deterministic decoders have normalized source dimension at most $41/64 + 5/(3\sqrt{d})$, over every field and with coupled entry lanes. For $M \le d/2$ classes, a difference-cover argument gives a deficit of order $M^{-1/2}$. A cyclic lift preserves a fixed class count and gives two-class binary sources of dimension $b(d^2/2 + 7d/6)$ on an infinite family of dimensions, with a paid two-work-tape realization at bit and canonical logarithmic widths. Complete composition has different constraints: two linear factors with bounded class counts cannot transpose in unbounded dimension. We prove quantitative rank and separately scoped nonlinear obstructions using balance, literal periods and equivariant outer maps. One shared decoder can also be correct on an almost full-entropy arbitrary set, showing why entropy alone cannot replace source linearity. These results strengthen the algebraic restrictions and clarify what a native execution would have to supply; they do not identify arbitrary tape computations with this model. All numbered results are Lean-verified except the native two-work-tape implementation theorem, which has a complete written proof and an explicit formalization todo.

Contributions. Hans Brende chose the problem and directed the research. GPT-6 Astra (OpenAI) developed the mathematics, Lean formalizations and text under his direction.

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1 Introduction

Matrix transposition exchanges row and column addresses while preserving every entry. The series studies whether a fixed deterministic multitape Turing machine can perform this operation in time linear in the bit length, the question (T) of [2]. A candidate computation must both move information and organize the local rules that use it. This paper isolates the second obligation in the cyclic shared-decoder model introduced in [1].

One rule can be correct on a large source without being correct everywhere. Nor does a large source for each stage guarantee that the stages compose to transposition. We therefore keep two questions separate. Part I bounds and constructs linear correctness sources for a single rule, allowing its decoders to be arbitrary deterministic functions. Part II studies exact complete-map factorizations, distinguishing linear factors from the nonlinear architectures for which additional symmetry arguments apply.

Results and their use. The two-class ceiling improves the earlier $3/4 + o(1)$ bound for linear decoders to $41/64 + o(1)$, now allowing arbitrary deterministic decoders. The proof intersects translated sources inside the smaller invariant spaces created by earlier intersections and retains a negative doubled-step term before averaging. For $M \leq d/2$, the many-class estimate gives an inverse-square-root deficit. These are quantitative advances on the arbitrary-class capacity question in [1, Problem 9.1(a)]; the half-rate target remains open. Both bounds restrict the linear sources a construction may use; neither is a native time lower bound.

On the construction side, a cyclic lift retains the number of receiver classes. It gives a growing additive advantage over the interval bound with two classes, while its normalized advantage tends to zero. The revised seventeenth paper already gives unbounded additive excess for every fixed number of classes, over every field, including $\mathbb{F}_2$[1]. Our contribution here is the general lift preserving the seed classes, its exact periodic-capacity formula, and the full rule kernel of its binary specialization; binary fixed-class excess itself is already known. A paid realization also separates a finite algebraic description from a native implementation. It remains correct only on its specified source.

The principal composition obstruction addresses the many-hop gap identified in [1, Section 7.4]. It excludes exact products of two complete linear rules with fixed class counts for all sufficiently large dimensions, over every field and with coupled lanes. Its common-message lemma and the rank-deficit extension explain the additional obligation imposed by a complete intermediate state. Nonlinear statements require different hypotheses: equivariant surrounding maps, actual commutation identities, or a specified first factor. No unrestricted nonlinear two-stage exclusion is claimed.

The extensions test which premises a native reduction must preserve. Even one shared decoder can work on an arbitrary correctness set with only logarithmic entropy deficit, while deterministic line rules with a fixed number of classes have asymptotic rate at most one half on linear sources. Thus extracting a large fibre does not by itself make the capacity theorem applicable. The exhibited cyclic lift's dimension is now its entire agreement-kernel dimension. An all-width reflection family and a sharper extension-linear residual theorem separate two different linearity obligations. A complete nonlinear first factor can satisfy the one-stage requirements yet force every class of its remaining factor to be a singleton. These statements narrow specific interfaces; none identifies the algebraic model with every native tape execution.

Proofs and formal status. A line marked Lean names the authoritative formal statement and proof; the English outline is a reading aid. A partial line identifies the exact missing step, and a written-only result has a complete written proof. Every numbered mathematical result is formalized except the native two-work-tape implementation in Theorem 5.4; its transition-level execution proof

remains a formalization todo. The checked results include orbit enumeration, finite probability estimates, the actual reflection/Fourier adapter, and complete residual accounting. All formal files use the shared libraries of this repository and Mathlib [7]; standard algebra, finite-field and Ramsey theorems are credited where used.

Organization. We first fix the common model and its two correctness notions. Part I develops capacity bounds, structural reductions and constructions. Part II turns to linear composition and nonlinear balance and period constraints. The final main section states the remaining mathematical and native-machine interfaces. For the source-linearity premise, read Appendix A; for decoder linearity, read Appendices D, F and G. The actual diagonal defects and mask-method limits refine the capacity search in Appendices B, C and E. The remaining appendices test complete first factors and balanced nonlinear maps. Finally Appendix K gives a conditional operational route: an exhibited complete affine baseline turns its operator defect into a lower bound on the joint nonlinear correction record.

2 The shared-decoder model

Let $d \geq 1$ and write $I = \mathbb{Z}/d\mathbb{Z}$. For an entry alphabet A, the complete state space is $\mathcal{X} = A^{I\times I}$. In the linear setting $A = F^b$, where F is a field and $b \geq 1$; thus $N = bd^2$ is the dimension over F. For binary tape inputs the bit length is $L = d^2b$. The canonical width in (T) is $b(d) = \max(1, \lceil \log_2 d \rceil)$, but algebraic statements will quantify over every positive b.

Define the column and row shifts and the transpose by

$$(T^s X)_{i,j} = X_{i,j+s}, \qquad (R^s X)_{i,j} = X_{i+s,j}, \qquad (\tau X)_{i,j} = X_{j,i}.$$

On a row or column word, σ^s shifts its single cyclic index in the same orientation. Thus $\tau T^s = R^s \tau$, $\tau R^s = T^s \tau$, and $T^s R^t = R^t T^s$.

Definition 2.1 (A shared rule and its correctness source)**.** `Lean`: `ClassUniform`, `IsSharedCode` [`Model.lean`]

A receiver partition $p : I \to \{1, \ldots, M\}$ uses every label. A complete shared rule $K : \mathcal{X} \to \mathcal{X}$ has total functions $f_J : \mathcal{X} \to A^I$ such that

$$\operatorname{col}_c K(X) = f_{p(c)}(T^c X).$$

The decoder has no additional receiver-index argument. In the linear setting, a subspace $C \leq \mathcal{X}$ is a correctness source if $f_{p(c)}(T^c X) = \operatorname{row}_c X$ for every $X \in C$ and every c. Its dimension is denoted D.

The predicates are distinct: a correctness source need not contain arbitrary intermediate states of a composition. A linear source does not imply that its decoders are linear. A *linear complete rule* means that K itself is F-linear on all of $\mathcal{X}$. All definitions permit mixing between lanes. A map H is column-equivariant if $HT^s = T^s H$ for every s; one-class complete rules are exactly these maps. Equivariance does not assert that the map is cheap to compute on tapes.

All complete factorizations in Part II act on this same state space and use this same column-view action, unless a theorem explicitly says otherwise. For a finite alphabet, exact factorization of the bijection τ by total square maps forces every factor to be bijective. Correctness on a proper source, or a larger workspace with a promised initialization, does not imply this conclusion. Algebraic row or column operations incur no native time charge until a separately stated implementation supplies it.

Part I
Capacity and constructions

3 Capacity of deterministic shared decoders

Throughout this section F is a field, $d, b \geq 1$, and $V = (F^b)^{(\mathbb{Z}/d)\times(\mathbb{Z}/d)}$ has dimension $N = bd^2$. A partition p of the receiver indices has M nonempty classes. The source $C \leq V$ is linear, of dimension D; the shared decoders $f_J : V \to (F^b)^d$ are arbitrary total functions satisfying $f_{p(c)}(T^c X) = \mathrm{row}_c\, X$ for $X \in C$. All lanes may be coupled. This distinction between a linear source and a linear decoder will remain explicit.

3.1 Intersect inside the covariance space

For a nonempty set $E \subseteq \mathbb{Z}/d$, put

$$C_E = \bigcap_{e\in E} T^{-e} C.$$

Join distinct indices c, c' when $p(c) = p(c')$ and $c' - c \in E - E$; write $\chi(E)$ for the number of components. Let H_E be the subspace defined by

$$\mathrm{row}_{c'}\, X = \sigma^{c'-c}\, \mathrm{row}_c\, X \quad \text{on every edge.}$$

Lemma 3.1 (Deterministic translate pigeonhole)**.** `Lean`: `IsSharedCode.component_finrank_le`, `finrank_covarianceSpace`, `covarianceSpace_rot` [`CapacityAmbient.lean`]
For every nonempty E, $C_E \subseteq H_E$, and $\dim H_E = bd\chi(E)$. Consequently

$$|E|D \leq (|E| - 1)N + bd\chi(E).$$

Every H_E is invariant under every column rotation T^t.

Outline of the formal proof. If $c + e = c' + e'$ with $e, e' \in E$ and $p(c) = p(c')$, apply the two correct decoder equations to $T^e X, T^{e'} X \in C$. Their arguments are the same full matrix, so $\sigma^e\, \mathrm{row}_c\, X = \sigma^{e'}\, \mathrm{row}_{c'}\, X$. This gives the edge equation without using linearity of the decoder. After replacing row c by $\sigma^{-c}\, \mathrm{row}_c\, X$, the equations say simply that normalized rows are constant on each component. One arbitrary vector of dimension bd per component gives all solutions; there is no further condition around a cycle. This proves the dimension. Codimension is subadditive under intersection, and each T^e is invertible, so $\dim C_E \geq |E|D - (|E| - 1)N$. Finally, applying the same rotation σ^t to all rows preserves every edge equation because the rotations commute. This proves invariance. □

The covariance argument is that of the translate-pigeonhole theorem in [1]. Its use of decoder values does not require those decoders to be linear. The smaller invariant ambient in the next theorem is the additional resource.

Theorem 3.2 (Iterated translate bound)**.** Lean: `IsSharedCode.iterated_finrank_le`, `componentCount_zero` [`CapacityIteration.lean`]

Choose shifts $t_1, \dots, t_k$, allowing repetitions, and set

$$E_0 = \{0\}, \qquad E_j = E_{j-1} \cup (E_{j-1} + t_j), \qquad n_j = bd\chi(E_j).$$

Then $n_0 = N$ and

$$D \le \sum_{j=0}^{k-1} \frac{n_j}{2^{j+1}} + \frac{n_k}{2^k}. \tag{1}$$

Outline of the formal proof. Write $C_j = C_{E_j}$. The set identity gives $C_j = C_{j-1} \cap T^{-t_j} C_{j-1}$. Both spaces on the right lie in $H_{E_{j-1}}$, by the preceding lemma and rotation invariance. They have the same dimension. Grassmann's formula inside this ambient gives $2 \dim C_{j-1} \le n_{j-1} + \dim C_j$. Iterate these inequalities and use $\dim C_k \le n_k$. Repeated subset sums are counted once in E_j; their collisions do not affect either the set identity or the argument. □

The theorem bounds actual simultaneous intersections. It assumes neither independence of the constraints nor correctness of H_E itself. It can be combined with every applicable direct translate bound.

3.2 A uniform bound for two classes

Suppose now that p has at most two labels. For $s \in \mathbb{Z}/d$, let $g_s = \gcd(d, s)$, where $g_0 = d$, and $Q_s = |\{c : p(c) \ne p(c+s)\}|$. On the cycles of addition by s, let $R_j(s)$ count proper monochromatic runs of length at least j. A proper run is preceded by a color change; a monochromatic cycle contributes no runs. Thus $R_1(s) = Q_s$.

Lemma 3.3 (Run components and doubled steps)**.** Lean: `componentCount_le_runs_add_two_gcd`, `componentCount_le_changes_add_gcd` [`CapacityRunsComponents.lean`]

Lean: `CapacityRuns.changeCount_double`, `CapacityRuns.weighted_run_budget` [`CapacityRuns.lean`]

For every integer $r \ge 1$, the same-color graph with edges at s-cycle distance at most r has at most $R_r(s) + 2g_s$ components. At radius one it has at most $Q_s + g_s$. For every s, including zero and steps with short cycles,

$$Q_{2s} = 2R_2(s), \tag{2}$$

$$Q_s + R_2(s) + 2R_4(s) + 2R_{13}(s) \le d. \tag{3}$$

Outline of the formal proof. Work on one s-cycle. If its length is at most $2r$, the radius-r graph is complete within each color and has at most two components. Otherwise list the positions of one occurring color cyclically. Consecutive positions are connected when their intervening gap is at most r. Every other short edge is a path of these consecutive short edges, so it adds no component merger. If $B > 0$ gaps are longer than r, there are B components; if no gap is long, there is one. A singleton has its one long cyclic gap and one component. For a nonconstant binary cycle, these long gaps are exactly the opposite-color runs of length at least r. Summing the two colors proves the first estimate. At radius one a nonconstant cycle has exactly one component per run; a monochromatic cycle adds one component and no change. Summing cycles proves both claims.

For the identity, write $z_c = 1_{p(c) \ne p(c+s)}$. Then $1_{p(c) \ne p(c+2s)} = z_c \operatorname{xor} z_{c+s}$. In a nonconstant, nonalternating cycle, the changes of z number twice its nonempty runs of ones, each ending at the start of a p-run of length at least two. Constant and alternating cycles give zero on both sides.

This also handles cycles of lengths one and two. Finally a proper run of length L contributes $1 + 1_{L\geq 2} + 2\,1_{L\geq 4} + 2\,1_{L\geq 13}$ to the left of (3). Its values are $1, 2, 4, 6$ on the successive ranges $L = 1$, $2 \leq L \leq 3$, $4 \leq L \leq 12$, $L \geq 13$; each is at most L. Monochromatic cycles contribute zero. Sum the run lengths. □

Theorem 3.4 (Theorem C: the universal two-class ceiling)**.** `Lean: IsSharedCode.two_class_capacity_bound` [`CapacityUniversal.lean`]

For every field, every $d, b \geq 1$, every partition into at most two classes, and every linear source with deterministic shared decoders,

$$D \leq \frac{41}{64}bd^2 + \frac{5}{3}bd^{3/2}. \tag{4}$$

In particular the normalized dimension is at most $41/64 + o(1)$, uniformly in the partition, field, word width and decoders.

Outline of the formal proof. Apply Theorem 3.2 with successive shifts $s, 3s, 9s$. The three difference sets contain every multiple of s with coefficient in $[-1, 1]$, $[-4, 4]$, and $[-13, 13]$, respectively: this follows by choosing difference digits from $\{-1, 0, 1\}$ in bases $1, 3, 9$. Reduction modulo d may identify elements, which causes no problem. The component estimates give

$$D \leq \frac{1}{2}bd^2 + \frac{1}{4}bdQ_s + \frac{1}{8}bd\big(R_4(s) + R_{13}(s)\big) + \frac{3}{4}bdg_s.$$

Use (3) and (2) to obtain

$$D \leq \frac{9}{16}bd^2 + \frac{3}{16}bdQ_s - \frac{1}{32}bdQ_{2s} + \frac{3}{4}bdg_s. \tag{5}$$

The negative term is retained before averaging.

Choose a uniformly from all d residues and put $A_r = \mathbb{E}_a Q_{2^r a}$ and $B_r = \mathbb{E}_a \gcd(d, 2^r a)$. If the label sizes are n_0, n_1, ordered-pair counting gives $A_0 = 2n_0n_1/d \leq d/2$. Average (5) at $s = 2^r a$, then weight these inequalities by $w_r = (5/6)6^{-r}$ for $r \geq 0$. The weights sum to one and all terms are bounded for fixed d. Since $(3/16)w_r = (1/32)w_{r-1}$ for $r \geq 1$, the internal A_r terms cancel, leaving

$$D \leq \frac{41}{64}bd^2 + \frac{3}{4}bd\sum_{r\geq 0} w_r B_r.$$

Equivalently, sum through a finite index and then let the index grow; the terminal term has absolute value at most $bd^2 w_r/32$ and tends to zero. No uniformity of $2^r a$ on the whole cycle is assumed.

Let $a_r = \gcd(d, 2^r)$ and $n = d/a_r$. Its actual distribution is uniform on the multiples of a_r, so $B_r = a_r \mathbb{E}_{u\in\mathbb{Z}/n} \gcd(n, u)$. For each divisor e of n, at most n/e residues have gcd equal to e. Thus the mean gcd is at most the number of divisors of n, which is at most $2\sqrt{n}$ by pairing divisors across $\sqrt{n}$. This includes $u = 0$ and $n = 1$. Consequently

$$B_r \leq 2\sqrt{d}\,2^{r/2}, \qquad \sum_{r\geq 0} w_r B_r \leq \frac{5\sqrt{d}}{3(1 - \sqrt{2}/6)} \leq \frac{20}{9}\sqrt{d}.$$

Substitution proves (4). The proof works with rows in $(F^b)^d$ throughout; no lane decomposition of C was used. □

The bound improves the general two-class $3/4 + o(1)$ ceiling for linear decoders stated in [1, Open problems], and allows arbitrary deterministic decoders. It leaves the conjectured $1/2 + o(1)$ ceiling open. The parameter being bounded is the dimension of a linear correctness source, not the density of native algorithms.

3.3 Arbitrarily many classes

Theorem 3.5 (Theorem M: a square-root class deficit)**.** Lean: `IsSharedCode.difference_cover_real_bound`, `IsSharedCode.square_cover_bound`, `IsSharedCode.saturated_cover_bound` [`CapacityDifferenceCover.lean`]

For positive integers u, v, set $R = \min(uv, d-1)$ and $U = \min(d, M\lfloor d/(R+1)\rfloor)$. Then

$$D \le N - \frac{bd(d-U)}{u+v}. \tag{6}$$

If $M \le d/2$ and $t = \lceil\sqrt{2M}\rceil$, then

$$D \le \left(1 - \frac{1}{4t}\right) N. \tag{7}$$

For every $d \ge 2$ and every M, writing $t_d = \lceil\sqrt{d-1}\rceil$, $D \le N - bd(d-M)/(2t_d)$.

Outline of the formal proof. Use the shift set $E = \{0, 1, \dots, u-1\} \cup \{u, 2u, \dots, vu\}$, reduced modulo d. Its size is at most $u+v$. Every integer $1 \le h \le uv$ equals $ju - i$ with $j = \lceil h/u\rceil \le v$ and $0 \le i < u$. Thus $E - E$ contains every signed difference through R.

For one class, list its vertices cyclically. The positive gaps sum to d, including the sole gap d for a singleton. Keep consecutive edges whose gap is at most R. If there are $\ell \ge 1$ larger gaps, there are at most ℓ components and $\ell(R+1) \le d$. If there is no large gap, there is one component, also at most $\lfloor d/(R+1)\rfloor$ because $R \le d-1$. Hence $\chi(E) \le U$. The translate bound gives $D \le N - bd(d-\chi(E))/|E|$. Both numerators are nonnegative; replacing $\chi(E)$ by U and $|E|$ by its upper bound proves (6).

For $u = v = t$, either $t^2 \ge d-1$, giving $U = M \le d/2$, or $U \le Md/(t^2+1) < d/2$. Equation (7) follows. For $u = v = t_d$ the radius is $d-1$ and $U = M$, giving the last claim. Reduction of the shift set may create repetitions; the proof uses only its upper cardinality bound. It includes $d = 1$ in the first inequality. □

This application uses the elementary sparse difference cover above, not a new difference-cover theorem. It changes the general deficit order from $1/M$ to $1/\sqrt{M}$, while stronger fixed-M bounds remain available. For example, if $D \ge (1-\varepsilon)N$ and $0 < \varepsilon \le 1/8$, then (7) gives the precise dichotomy

$$M > d/2 \quad \text{or} \quad M > 1/(128\varepsilon^2).$$

Indeed $\lceil\sqrt{2M}\rceil \ge 1/(4\varepsilon)$ forces $\sqrt{2M} > 1/(4\varepsilon) - 1 \ge 1/(8\varepsilon)$. The saturation at d is necessary because singleton classes can decode the full source. Nothing here extracts these decoder interfaces from a native tape execution or closes them under composition.

4 Completion and exact interval capacity

4.1 A normal form without loss of dimension

Choose a representative r_J from each class J. Define

$$(PX)_c = \sigma^{c-r_J} \operatorname{row}_{r_J} X \quad (c \in J),$$
$$H_* = \operatorname{im} P, \qquad W = \ker P = \{X : \operatorname{row}_{r_J} X = 0 \text{ for every } J\}.$$

Here P acts on rows, and each row has bd scalar coordinates. The class-covariant space H_* was already used in the small-code constructions of [1]. What matters here is that one may require its inclusion when maximizing capacity.

Theorem 4.1 (No-loss completion)**.** Lean: `IsSharedCode.completedCode`, `finrank_completedCode_add_rank`, `finrank_le_completedCode`, `completedCode_eq_of_range_le` [`CapacityCompletion.lean`]

The map P is an idempotent linear map commuting with every T^t, and $\dim H_* = Mbd$. If C is a linearly supported code with deterministic shared decoders, then

$$C^+ = (C \cap W) \oplus H_*$$

has deterministic shared decoders for the same partition and

$$\dim C^+ = \dim C - \operatorname{rank}(P|_C) + Mbd \geq \dim C.$$

If $H_* \subseteq C$, then $C^+ = C$ as subspaces.

Outline of the formal proof. The representative rows are unchanged by P, freely parametrize its range, and vanish exactly on its kernel. These facts prove idempotence, the range dimension, and the direct decomposition. Row rotations commute, so $PT^t = T^tP$. Given original decoders f_J, use

$$f_J^+(Y) = f_J((I-P)Y) + \sigma^{-r_J} \operatorname{row}_{r_J} Y. \tag{8}$$

For $X_0 \in C \cap W$, $H \in H_*$ and $c \in J$, the first argument in (8) at $T^c(X_0 + H)$ is T^cX_0. The second term is $\operatorname{row}_c H$, because the representative rows of X_0 vanish. Thus the output is the demanded row of $X_0 + H$. Rank-nullity for $P|_C$ gives the exact dimension. If $H_* \subseteq C$, then $PX \in C$ for $X \in C$, giving $C = (C \cap W) \oplus H_*$. □

The completion need not contain the original code: it replaces it, and may change every decoder away from required views. The residual is $C \cap \ker P$, not $(I-P)C$. If the original decoders are linear, (8) is linear as well.

Corollary 4.2 (Residual capacity and transversal rank)**.** Lean: `shared_capacity_eq_residual`, `linear_capacity_eq_residual`, `capacity_transversal_deficiency`, `optimal_transversal_full_rank` [`CapacityOptimization.lean`]

In either the deterministic-decoder or linear-decoder model, let $\operatorname{Cap}(p)$ be the maximum code dimension, and let $\operatorname{Cap}_W(p)$ be the maximum among codes contained in W. Then

$$\operatorname{Cap}(p) = Mbd + \operatorname{Cap}_W(p).$$

For every code C and every representative transversal,

$$\operatorname{Cap}(p) - \dim C \geq Mbd - \operatorname{rank}(P|_C).$$

In particular every dimension-optimal code projects onto the entire tuple of representative rows, for every such transversal.

Outline of the formal proof. Attainable dimensions form nonempty subsets of $\{0, \ldots, N\}$, so the maxima exist. For any C, rank-nullity gives $\dim C \leq Mbd + \dim(C \cap W) \leq Mbd + \operatorname{Cap}_W(p)$. Conversely, complete a residual maximizer. Completion preserves linearity when required, so both models obey the same equality. Applying maximality to the exact dimension of C^+ proves the displayed inequality. Row evaluation on the transversal has the same rank as $P|_C$; at a maximum that rank must be Mbd. □

4.2 Normal bases and disjoint residual views

We describe a cyclic-interval partition by a transversal containing every class boundary: there is one representative $r_J \in J$ per class, and every c with $p(c-1) \neq p(c)$ is one of these representatives. For a single class any representative may be used. This formulation includes intervals that cross the chosen coordinate origin.

Theorem 4.3 (Extremal interval codes)**.** Lean: `CapacityIntervals.exists_binary_code_dimension`, `CapacityIntervals.exists_optimal_interval_code`, `IsSharedCode.two_mul_finrank_le_interval` [`CapacityIntervals.lean`]

Let a partition of $\mathbb{Z}/d$ have M nonempty classes and let $d - M$ be even. There is a binary linear source of dimension

$$\frac{d^2 + Md}{2}$$

with deterministic shared decoding for this partition. For cyclic-interval classes this dimension is optimal. Repeating the construction in b lanes gives the corresponding dimension $b(d^2 + Md)/2$ for every $b \geq 1$.

Outline of the formal proof. Let $E = \mathbb{F}_{2^d}$ and $\Phi(x) = x^2$. By the normal-basis theorem [6, Theorem 5.18], choose α whose conjugates form a binary basis, and use its reversed order: $e_j = \Phi^{-j}\alpha$. The identification $\iota(x) = \sum_j x_j e_j$ satisfies

$$\iota(\sigma x) = \Phi(\iota(x)).$$

Indeed replacing j by $j + 1$ in the sum proves this identity with the stated rotation convention. Put $\beta = \alpha$. Its d Frobenius conjugates are distinct, since the basis elements are distinct.

Write $d - M = 2s$ and pair the nonrepresentative rows as (a_i, b_i). Define a residual source C_0 by assigning to each pair the rows $\iota^{-1}(x_i), \iota^{-1}(\beta x_i)$, with independent $x_i \in E$; set every representative row to zero. Then $\dim C_0 = sd$. In $T^c C_0$ the same pair has the form $(u_i, \Phi^c(\beta)u_i)$. If a matrix belongs to both $T^c C_0$ and $T^e C_0$, $c \neq e$, equality of the two rows gives $(\Phi^c\beta - \Phi^e\beta)u_i = 0$ for every i. Thus all $u_i = 0$.

Consequently the nonzero translated views of C_0 are disjoint. Define one total function f on full matrices by $f(0) = 0$ and $f(Y) = \mathrm{row}_c(T^{-c}Y)$ if $0 \neq Y \in T^c C_0$; set it to zero outside these views. The phase c is unique, so this is well-defined and decodes all residual receivers. Complete C_0 by Theorem 4.1. Its dimension becomes $Md + sd = (d^2 + Md)/2$. This also covers $s = 0$, when the residual is zero.

For an interval partition, the graph in Lemma 3.1 with $E = \{0, 1\}$ has one component per interval. Hence $2 \dim C \leq d^2 + Md$. The case $d = 1$ is immediate. Independent lane copies give the asserted all-width construction, while the same deterministic translate bound gives its all-width interval upper bound. □

The normal-basis input is classical; it holds in even and composite degrees too. No primitive normal basis, root of unity, or diagonalization of the binary rotation is needed. The decoder identifies a phase by examining a full residual view. This is a construction in the shared-view model, without a tape implementation of that access.

At $d = 5$, classes $\{0\}, \{1\}, \{2, 3, 4\}$ have deterministic capacity 20. The comparison with the reported linear-decoder maximum 19 in [1] retains that source's solver qualification. The theorem proves the deterministic side and its upper bound, not a new formal proof of the reported linear nonexistence.

4.3 Why the hypotheses stay separate

Proposition 4.4 (Deterministic decoding need not linearize)**.** Lean: `CapacityControls.exists_deterministic_not_linear` [`CapacityControls.lean`]

There is a one-class binary linear source with deterministic shared decoding and no globally linear shared decoder.

Outline of the formal proof. At $d = 3$ let X have row zero $(1, 1, 0)$ and its other rows zero, and take $C = \mathrm{span}_{\mathbb{F}_2}\{X\}$. The three nonzero views X, TX, T^2X are distinct, so a deterministic map may send them respectively to $(1, 1, 0), 0, 0$, and send zero to zero. Those are exactly the demanded rows. But $X + TX + T^2X = 0$, while the required outputs have nonzero sum. No linear function has these values. □

Proposition 4.5 (A high-dimensional affine marker)**.** Lean: `CapacityMarker.exists_affine_source_decoder` [`CapacityMarker.lean`]

Over every field and for every $d \geq 1$, a one-class deterministic shared decoder transposes an affine source of dimension $d^2 - d$.

Outline of the formal proof. Require row zero to be the coordinate word e_0 and leave all other entries free. The first row of T^cX is $\sigma^c e_0$, which identifies c uniquely. A total decoder reads that marker, applies T^{-c}, and returns row c; define it arbitrarily on other full views. This proves correctness on all matrices in the affine source and its dimension. For $d > 3$ this exceeds the one-class linear-source bound $(d^2 + d)/2$, so a linear-source hypothesis cannot be replaced by an affine-dimension assertion for arbitrary deterministic decoders. □

Both controls concern complete shared views. The marker is not an unpaid row-addressing operation for a tape machine, and the strictness example does not compare the maximum capacities of two decoder models.

5 Cyclic lifts with the same classes

The original blow-up construction in [3] gives unbounded absolute excess over the interval bound with a growing number of classes. Its revised memo also gives a two-class parity-coset family for every $d = 4k \geq 8$ over fields where $2 \neq 0$ [3, Theorem 4.8]. The revised seventeenth paper extends that rule to every field, including characteristic two, and uses mixed blow-ups to obtain unbounded excess for every fixed number of classes[1]. The construction below instead gives a lift for an arbitrary seed with the same number of classes, an exact periodic-capacity formula, and a paid realization of its exhibited binary rule. These fixed-class families all have vanishing normalized excess; fixed-class or binary additive excess alone is not a new conclusion here.

Theorem 5.1 (Deterministic cyclic lift)**.** Lean: `cyclic_lift` [`ConstructionLinearLift.lean`]

Let a linear source $C_s \leq (F^b)^{s \times s}$ have dimension D and deterministic shared decoders for an M-class partition p of $\mathbb{Z}/s$. Let $m \geq 1$ be coprime to s, with $m \neq 0$ in F, and put $d = ms$. The lifted partition $p^{\uparrow}(c) = p(c \bmod s)$ admits a linear source of dimension

$$D^{\uparrow} = \frac{b(d^2 - ms^2)}{2} + mD \tag{9}$$

with deterministic shared decoders. Linear seed decoders give linear lifted decoders.

Outline of the formal proof. Use Chinese-remainder coordinates $(a, r) \in \mathbb{Z}/m \times \mathbb{Z}/s$ for both rows and columns. For $z \in \mathbb{Z}/m$ define

$$Z_z(X)_{r,c} = \frac{1}{m} \sum_{a \in \mathbb{Z}/m} X_{(a,r),(z-a,c)}, \qquad (EX)_{(a,r),(h,c)} = Z_{a+h}(X)_{r,c}.$$

The map E is an idempotent. Its image consists of arrays whose outer indices enter only through their sum, and is parametrized by the m independent seed matrices Z_z. Thus $\operatorname{rank} E = mbs^2$. It commutes with column rotation and transpose. On its image transpose acts as the inner $s \times s$ transpose on each Z_z.

Take the direct sum

$$C^{\uparrow} = (\ker E \cap \operatorname{Fix} \tau) \oplus \{X \in \operatorname{im} E : Z_z(X) \in C_s \text{ for every } z\}.$$

The second summand has dimension mD. The symmetric space has dimension $bd(d+1)/2$ in every characteristic. Its image under E is exactly the space of m symmetric seed matrices, of dimension $mbs(s+1)/2$: commutation proves inclusion, and every such image vector is already symmetric and fixed by E. Rank-nullity gives the first summand's dimension $b(d^2 - ms^2)/2$.

For an arbitrary full view Y, define one decoder per original class by

$$F_J(Y)_{a,r} = (Y - EY)_{(a,r),(0,0)} + f_J(Z_a(Y))_r. \tag{10}$$

At a receiver with CRT index (u, v), literal rotation satisfies $Z_a(T^{(u,v)}X) = T_s^v Z_{a+u}(X)$. Thus the complete induced map is

$$K^{\uparrow}(X)_{(a,r),(u,v)} = (X - EX)_{(a,r),(u,v)} + f_{p(v)}(T_s^v Z_{a+u}(X))_r.$$

For a codeword, the first term is its symmetric part, and the second transposes its projected part. This proves correctness. The receiver index is not an extra argument to (10); only its seed class selects the function. In particular a nonlinear decoder is applied after averaging the seed matrix, never distributed through that average. All displayed operations are linear when the seed decoder is linear. □

For linear rules there is also an operator description. On the outer tensor square put $E = m^{-1} \sum_{t \in \mathbb{Z}/m} G^t \otimes G^{-t}$, where G is rotation. After regrouping outer and inner factors, the complete lifted rule is

$$K^{\uparrow} = (I - E) \otimes I + E \otimes K_s. \tag{11}$$

If a seed term has column factor $P_J S^e$, its lifted column factor is $G^{-t} \otimes P_J S^e = P_{J^{\uparrow}} \sigma_d^u$, where CRT gives $u \equiv -t \pmod m$ and $u \equiv e \pmod s$. Hence this is one allowed same-class operator, not a composition with a free coordinate permutation. If the seed code is the entire agreement kernel, the displayed lifted code is the entire kernel of (11).

Corollary 5.2 (A fixed two-class excess)**.** Lean: `binaryCyclicCode_linear`, `binaryCyclicPartition_surjective`, `finrank_binaryCyclicCode`, `binaryCyclicCode_interval_excess` [`ConstructionSeed.lean`]

For every $m \geq 1$ coprime to six, $d = 6m$ admits a binary two-class linear rule and a linear source of dimension

$$\frac{d^2}{2} + \frac{7d}{6}.$$

Its excess over the two-interval bound is $d/6$.

Outline of the formal proof. Use the published six-dimensional seed with classes $\{0,1,3\},\{2,4,5\}$ and its verified 25-dimensional code from [1, 3]. Denote the complete rule induced by these seed decoders by K_6. Since m is odd, it is invertible in $\mathbb{F}_2$. Equation (9) gives $18m^2+7m = d^2/2+7d/6$; subtract $d^2/2+d$. The conclusion needs a seed subcode of dimension 25, not a proof that 25 is the seed's maximum capacity. □

5.1 The exact linear capacity of a periodic lift

Write $\mathrm{Cap}_F^{\mathrm{lin}}(p)$ for capacity with linear decoders over F.

Theorem 5.3 (Periodic capacity reduction)**.** Lean: `periodic_linear_capacity_split`, `periodic_linear_capacity_lower`, `periodic_linear_capacity_algebraicClosure_upper` [`ConstructionCapacity.lean`]

In the scalar case $b=1$, suppose additionally that F contains all m-th roots of unity. Then

$$\mathrm{Cap}_F^{\mathrm{lin}}(p^{\uparrow}) = \frac{d^2-ms^2}{2} + m\,\mathrm{Cap}_F^{\mathrm{lin}}(p).$$

Without the splitting assumption, the same expression with F is a lower bound, and the expression with its algebraic closure $\overline{F}$ is an upper bound.

Outline of the formal proof. The lower bound is Theorem 5.1. For the upper bound, the allowed column space in CRT coordinates is $\mathrm{Circ}_m \otimes \mathrm{span}\{P_J S^e\}$. Simultaneously diagonalize the outer circulants, using the same basis on both tensor factors so that the demand remains swap. This is valid because m is invertible and its roots split. Write the input as an $m \times m$ array of $s \times s$ blocks X_{ij}. Correctness has the form

$$X_{ji}^{\mathsf{T}} = \sum_a L_{ia}^{j}(X_{aj}),$$

where each L_{ia}^{j} is an allowed seed rule. In particular $\delta_j = \dim\ker(\tau_s - L_{jj}^{j}) \le \mathrm{Cap}_F^{\mathrm{lin}}(p)$.

Supply all upper-triangular blocks X_{ij}, $i<j$, and process j in decreasing order. All off-diagonal blocks of column j are known: those above were supplied, those below belong to recovered rows. The equation with $i=j$ determines X_{jj} up to a vector in a fixed δ_j- dimensional kernel. Supply δ_j linear coordinates injective on that kernel; this determines X_{jj} uniquely even on an affine fibre. The other equations now recover row j. This is an injective linear description of size $s^2m(m-1)/2 + \sum_j \delta_j$, proving the bound. For nonsplitting F, embed a basis of a maximizing source into $\overline{F}$ and take its $\overline{F}$-span. Scalar extension preserves the basis's independence and extends every class rule by its coordinate coefficients. The extended source has the same dimension and the same receiver classes. Apply the split result over $\overline{F}$. □

Thus every rule on a fixed periodic seed partition has rate $1/2 + O(1/d)$, even if it is not the exhibited lift. The qualitative limit already follows from the period obstruction in [1]; the equality records the exact finite seed-capacity contribution.

5.2 A paid implementation of the exhibited binary lift

Theorem 5.4 (Two-tape implementation)**.** Not formalized; a complete proof is given below.

There is a fixed finite-alphabet, finite-control machine with two initially blank work tapes, read-only input and append-only unreadable output that computes the binary operator in Corollary 5.2 lane by lane in $O(P)$ time, $P = d^2 b$, on admitted dimensions $d = 6m$, $\gcd(m, 6) = 1$. There are separate machines for $b = 1$ and for $b = \max(1, \lceil \log_2 d \rceil)$. Each has a transpose-correct subcode of dimension $b(d^2/2 + 7d/6)$.

Proof (not formalized). Here is a complete bounded-state schedule, including preparation. Put $\mu = mb$. The first work tape has m consecutive b-cell words and 36 binary payload tracks per physical cell. For $z \in \mathbb{Z}/m$ their intended contents are

$$H_z[i, j] = \bigoplus_{r \equiv i \ (6),\ c \equiv j \ (6),\ r+c \equiv z \ (m)} X_{r,c}, \qquad 0 \le i, j < 6.$$

Each word has independent FIRST/LAST marker bits; HOME is at cell zero, END at μ, and START at one word's first cell. Coincident markers when $b = 1$ or $m = 1$ are distinct alphabet tracks. All payload starts zero. After preparation the second work head parks throughout both data passes.

At the start of input row r, START and the bank head are at word $r \bmod m$, with finite registers $i = r \bmod 6$, $j = 0$, and lap count zero. A DATA step reads the old raw bit and old bank symbol, advances both heads right, and either toggles $H[i, j]$ by that bit (pass one) or appends $x + H[i, j] + (K_6 H)[i, j]$ (pass two), preserving the bank. The last expression is a fixed Boolean function of 36 old bits. An OLD LAST marker increments j modulo six and enters LAND. LAND at END rewinds physically to HOME; else it is at a FIRST marker. Returning to START counts a completed lap. Six laps finish one row. Importantly j is not reset at a lap boundary: m need not be divisible by six. At row end clear START, physically advance one whole word, wrapping at END if necessary, and set START there; then increment i modulo six and reset j and the lap counter. Perform this offset after the last row too. Only the row-start mode tests the raw endmarker; housekeeping preserves payload, holds raw still, and emits nothing.

Induction gives bank address $((r + c) \bmod m)b + t$ at lane t of entry (r, c), with $j = c \bmod 6$. Thus pass one computes exactly H. After d rows its START is HOME. Rewind raw across its entire input, preserving the bank, and run the same schedule in emission mode. There are exactly P appends in ordinary row-major order. Because m is odd, the sum in H is precisely the binary averaging projector E; the emitted formula is (11). All reads are pretransition reads, and all writes are at the old head locations.

For completeness one pass has P DATA steps, d^2 ordinary LAND steps, and six physical END wraps per row, costing $6d(\mu + 1)$ extra steps. Row offsets cost $db + d + 6(\mu + 1)$, and row-start branches cost d. The total is

$$2P + d^2 + 2db + 8d + 6 \le 5P \qquad (d \ge 6, b \ge 1).$$

The between-pass raw rewind costs $P + 2$; the final absent transition halts.

It remains to establish this bank from blanks without size advice. Count the raw length on the second tape using a little-endian binary counter and a marked least-significant cell. A carry through k ones and return costs $O(k + 1)$, whose total over P increments is $O(P)$; hold raw fixed during each carry, then advance it. Rewind raw. Fixed schoolbook arithmetic on a constant number of delimited $O(\log(P + 2))$-bit registers verifies $P = d^2$ or $P = d^2 \max(1, \lceil \log_2 d \rceil)$ by binary search, and checks $d = 6m$, $\gcd(m, 6) = 1$. These functions are strictly increasing in d, and all arithmetic takes $\mathrm{polylog}(P + 2)$ time. Reject other lengths before output.

The first head now writes the bank sequentially while the second head maintains counters for m words and b cells. Binary decrement and return has aggregate $O(b + \log(P + 2))$ cost per word; no full-width zero test is performed per cell. Setup, copying and register trips cost at most polylog$(P + 2)$ per word. Install all markers, write END, return to HOME, and erase the second tape's delimited scratch frame. This costs $O(P + m \log^c(P + 2) + mb + \log^c(P + 2)) = O(P)$ for a fixed exponent c, since $m = d/6$ and $d \log^c(d+2) = O(d^2 b)$ in both width regimes. Every unbounded counter was tape data, and all other registers range over fixed finite sets. This establishes the promised initial boundary with exactly two work heads and proves the uniform time bound. □

This is an implementation of one displayed operator on all inputs of its admitted lengths. Its transpose-correct source has rate tending to one half, so it is not a full-input transposer or a lower bound against native machines. No general class-uniform operator is assigned a free implementation.

Part II
Composition and nonlinear obstructions

6 Two linear stages

A class bound for one stage is not closed under composition: even two small-class maps can have many different endpoint rules. We therefore keep both complete operators. Their exact composition forces a *common message* on a small rotation-invariant space. Arithmetic progressions then force this message space to grow.

6.1 One message on several translates

Lemma 6.1 (Common-message growth)**.** Lean: `commonMessage_finrank_le` [`CompositionProgression.lean`]

Let A be a finite-dimensional vector space over any field, let $R\colon A \to A$ be linear, and let $W \le A$ have dimension D. Suppose that a linear map $L\colon A \to Z$ is injective on W and

$$L(R^j x) = L(x) \qquad (x \in W,\ 0 \le j < \ell),$$

where $\ell \ge 1$. If $f = \dim \ker(R - I)$, then

$$\ell D \le \dim A + (\ell - 1) f. \tag{12}$$

Invertibility of R is not required.

Outline of the formal proof. Put $V_t = W + RW + \cdots + R^{t-1}W$ and $V_0 = 0$. For $t \le \ell$, the map L vanishes on $(R - I)V_{t-1}$. This image is disjoint from W, because L is injective on W, and their sum lies in V_t. Rank–nullity therefore gives

$$\dim V_t \ge D + \dim V_{t-1} - f.$$

Starting with $V_1 = W$ and using $\dim V_\ell \le \dim A$ proves the claim. In fact telescoping gives $V_t = W \oplus (R - I)V_{t-1}$, but the formal proof only needs inclusion. □

The correction f matters. Over $\mathbb{F}_2$, take a three-coordinate rotation and $W = \operatorname{span}\{(1,1,1), e_0\}$. Projection along $\operatorname{span}\{(R-I)e_0\}$ identifies W and RW with the same two-dimensional message. Thus $\ell D = 4 = 3 + 1$ for $\ell = 2$; a bound without the fixed-space term would be false.

For positive integers M, ℓ, write $\mathsf{W}(M,\ell)$ for any finite interval length such that every M-coloring of $\{0,\dots,\mathsf{W}(M,\ell)-1\}$ has a monochromatic ordinary ℓ-term arithmetic progression with positive step. Existence is the finite van der Waerden theorem. Our formal library derives this version from Mathlib's finite Hales–Jewett theorem [4]: summing the symbols of a combinatorial line gives an ordinary progression in a fixed bounded interval (`exists_progression_bound`). We use no numerical estimate for W.

Theorem 6.2 (Vanishing common-message rate)**.** Lean: `commonMessage_interval_bound` [`CompositionChannels.lean`]

Let $A = (F^q)^{\mathbb{Z}/d}$, with simultaneous cyclic rotation T, where $q, d \geq 1$ and F is any field. Let $W \leq A$, and let $B\colon W \to Z$ be linear and injective. Suppose the d positions have M nonempty classes and each class J has a linear map $L_J\colon A \to Z$ such that

$$L_J(T^c x) = B(x) \qquad (c \in J,\ x \in W).$$

For $\ell \geq 2$, put $N_0 = \mathsf{W}(M,\ell)$. If $d \geq N_0$, then

$$\dim W \leq \frac{q(d + N_0 - 1)}{\ell}. \tag{13}$$

Outline of the formal proof. Choose a monochromatic progression $c_0, c_0 + s, \dots, c_0 + (\ell-1)s$ among the first N_0 positions. Here $s \geq 1$ and $(\ell-1)s \leq N_0 - 1$. Apply Lemma 6.1 with $R = T^s$ and $L = L_J T^{c_0}$. A word fixed by T^s is determined by its first s coordinates, by repeatedly reducing a natural-number index by s. Thus the fixed space in A has dimension at most qs over every field. No diagonalization is used. Thus

$$\ell \dim W \leq qd + (\ell-1)qs \leq q(d + N_0 - 1).$$

□

For fixed M, this implies $\dim W/(qd) \to 0$ uniformly in F, q, the partition and the linear maps: choose ℓ first, then take $d \geq \mathsf{W}(M,\ell)$, so the ratio is less than $2/\ell$.

Linearity here concerns the decoder, not merely the source. To see the distinction, identify binary words of length d with $E = \mathbb{F}_{2^d}$ in a normal basis, so rotation is Frobenius Φ [6]. Choose β with d distinct conjugates. The binary-linear two-channel space $W = \{(x, \beta x) : x \in E\}$ has dimension d, and its distinct rotated views intersect only at zero. A single total nonlinear decoder sends $(u, \Phi^c(\beta)u)$, for $u \neq 0$, to $\Phi^{-c}u$ and sends all other points to zero. It recovers the same original x at every view with rate $1/2$. This is a source-level counterexample to removing decoder linearity; it is not a two-stage transpose factorization.

6.2 The complete two-stage reduction

We now use the matrix model with b coordinates per entry, allowing arbitrary mixing of rows and lanes by every linear decoder. Both stages use the same column rotation. Their partitions need not agree.

Theorem 6.3 (Two linear stages)**.** Lean: `twoHop_progression_gcd`, `twoHop_coprime_progression_dimension_le`, `twoHop_dimension_le`, `exists_twoHop_cutoff` [`CompositionLinear.lean`]

Let K_1, K_2 be linear class-uniform maps on $F^{d\times d\times b}$, with M_1, M_2 nonempty classes, and suppose $K_2K_1 = \tau$ on the whole space. If a first-stage class contains $c_0, c_0+s, \dots, c_0+(\ell-1)s$, with $\ell \geq 1$, integer $s \geq 0$ and indices read modulo d, then

$$\ell d \leq M_2 d + (\ell - 1)M_2 \gcd(d, s). \tag{14}$$

In particular, a progression of length $M_2 + 1$ with $\gcd(d, s) = 1$ forces $d \leq M_2^2$. Without the coprimality premise,

$$d \leq M_2(\mathsf{W}(M_1, M_2 + 1) - 1). \tag{15}$$

Thus fixed bounds on both class counts exclude exact two-stage transposition at all sufficiently large d, over every field and uniformly over every positive width b.

Outline of the formal proof. Both square factors are invertible, and $K_1^{-1} = \tau K_2$. Let $S_0 = \mathrm{Fix}(T)$, of dimension bd, and let $P_2 \leq F^d$ be the span of the second-class indicator vectors. Their disjoint nonempty supports give $\dim P_2 = M_2$ over every field. For $X \in S_0$, all views T^cX coincide. Hence K_2X is constant on each second receiver class. It follows that

$$W := K_1^{-1}S_0 \;\leq\; A := P_{2,\mathrm{row}} \otimes F^d_{\mathrm{column}} \otimes F^b, \qquad \dim W = bd, \qquad \dim A = bM_2d.$$

This is an actual consequence of the global product identity, not an independently chosen source. The ambient A is T-invariant and consists of bM_2 cyclic channels.

The common column of K_1X, for $X \in W$, defines one injective linear message $B(X)$. At every receiver c in a first class J, $f_{1,J}(T^cX) = B(X)$. Apply Lemma 6.1 on A to the specified progression. A word fixed by the s-step rotation is constant on its $\gcd(d, s)$ position orbits. Restriction to one representative per orbit is injective, so the fixed-space term is at most $bM_2 \gcd(d, s)$. The formal proof uses the actual translation-orbit quotient. Canceling $b > 0$ gives Eq. (14); putting $\ell = M_2 + 1$ and $\gcd(d, s) = 1$ gives $d \leq M_2^2$.

Put $N_0 = \mathsf{W}(M_1, M_2 + 1)$. If $d > M_2(N_0 - 1)$, then $d \geq N_0$. A monochromatic $(M_2 + 1)$-term progression in the first N_0 positions has positive step s with $M_2 s \leq N_0 - 1$. Since $\gcd(d, s) \leq s$ for this positive step, (14) would imply

$$d \leq M_2^2 s \leq M_2(N_0 - 1),$$

a contradiction. This proves the finite cutoff. Using fixed upper bounds for the numbers of colors gives one cutoff for all the stated parameters. □

The orbit correction can exclude particular masks before the general Ramsey threshold applies. For example, with two second-stage classes, a monochromatic three-term progression with step coprime to d already forces $d \leq 4$. The global two-class cutoff below remains 16, since Ramsey extraction does not require its progression step to be coprime.

The theorem concerns two complete maps on the same state space and tensor side. It neither multiplies endpoint class counts nor excludes additional uniform postprocessing. The familiar two partial-digit swaps at $d = M_1M_2$ remain valid controls. The bound does not prove that their product condition is necessary for arbitrary coded factors.

Corollary 6.4 (Two classes at each stage)**.** Lean: `twoHop_two_classes_le_sixteen` [`CompositionLinear.lean`]

If each of the two linear stages uses at most two classes, exact transposition requires $d \le 16$, over every field and every positive width.

Outline of the formal proof. The finite statement $\mathsf{W}(2,3) \le 9$ is checked by Lean's kernel over all colorings of nine positions. Its declaration is `binary_progression_nine_finite`; no minimality assertion is needed. Substitute this bound in Eq. (15). □

6.3 One global rank error

The preceding witness has dimension only bd. To obtain more than a row-scale obstruction, retain a whole band of frequencies and charge the same global error only once.

Lemma 6.5 (Recurrence growth)**.** Lean: `recurrence_finrank_le` [`CompositionRecurrence.lean`]

Let $R\colon A \to A$ and $L\colon A \to Z$ be linear on a finite-dimensional space, let $W \le A$ have dimension D, and let $k, m \ge 1$. Let $p(z) = \sum_{i=0}^{k} a_i z^i$. Suppose $(L, LR, \dots, LR^{k-1})$ is injective on W and

$$LR^j p(R)w = 0 \qquad (w \in W,\ j \ge 0,\ j + 2 \le mk).$$

Then, with $f = \dim \ker p(R)$,

$$mD \le \dim A + (m-1)f. \tag{16}$$

Outline of the formal proof. Set $V_t = \sum_{0 \le j < t} R^j W$ and $G = (L, \dots, LR^{k-1})$. For $0 \le t \le (m-1)k$, every component of G vanishes on $p(R)V_t$: its generator exponent is at most $t + k - 2 \le mk - 2$. Consequently $W \cap p(R)V_t = 0$. The degree bound gives $W + p(R)V_t \subseteq V_{t+k}$, so

$$\dim V_{t+k} \ge D + \dim V_t - f.$$

Start with $\dim V_k \ge D$ and use $t = k, 2k, \dots, (m-1)k$. Then $\dim V_{mk} \ge mD - (m-1)f$, proving the result. When $m = k = 1$ the displayed annihilation range is empty, as required. □

Theorem 6.6 (Growing two-stage rank deficit)**.** Lean: `twoHop_rank_interval_any_field`, `exists_twoHop_rank_threshold`, `twoHop_rank_superlinear` [`CompositionRankField.lean`]

Let the characteristic of F not divide d, and let K_1, K_2 be linear shared-decoder maps with M_1, M_2 classes on $F^{d \times d \times b}$. They need not be invertible. Put $r = \operatorname{rank}(\tau - K_2 K_1)$. For $k \ge 1$, set

$$\ell = (2M_2 + 1)k - 1, \qquad N_0 = \mathsf{W}(M_1, \ell).$$

If $d \ge N_0$, then

$$r \ge \frac{bk(d - N_0 + 1)}{2}. \tag{17}$$

For fixed class bounds, $r/(bd) \to \infty$ uniformly in the field, width, partitions and maps within this characteristic regime.

Outline of the formal proof. Extend scalars to a cyclotomic field with a primitive dth root ζ. The complete maps and their receiver classes extend, and the error rank does not increase: its extension factors through the scalar extension of its original range. It therefore suffices to prove the bound there. The characteristic hypothesis makes the roots distinct. Let $C = \ker(\tau - K_2 K_1)$ and $Y = K_1 C$. The restriction of K_1 to C is injective, since $K_1 x = 0$ there implies $\tau x = 0$. Thus $\dim Y \ge bd^2 - r$, with r denoting the original error rank.

Let S be the band of k consecutive frequencies, consisting of

$$y_{:,c,:} = \sum_{\nu=0}^{k-1} \zeta^{\nu c} a_\nu.$$

Its dimension is bdk, because $d \geq N_0 \geq \ell \geq k$. Pull $Y \cap S$ back through the injective restriction of K_1, obtaining $W \leq C$ of dimension $D \geq bdk - r$. This intersection charges the single global error r once.

Let $H \leq E^d$ be spanned by $c \mapsto 1_{p_2(c)=J}\zeta^{\nu c}$, over second classes J and $0 \leq \nu < k$, and put $h = \dim H \leq M_2 k$. On S, the second operator produces columns of the form $\sum_\nu \zeta^{\nu c} a_{p_2(c),\nu}$. Since $w = \tau K_2 K_1 w$ on W, we have

$$W \leq A := H_{\text{row}} \otimes E^d_{\text{column}} \otimes E^b, \qquad \dim A = bhd.$$

This ambient is invariant under column rotation.

Choose a monochromatic ℓ-term progression with step $s \geq 1$ among the first N_0 positions. On A, put $R = T^s$ and $L = f_{1,J} T^{c_0}$. For $w \in W$,

$$LR^j w = \sum_{\nu=0}^{k-1} \zeta^{\nu c_0} (\zeta^{\nu s})^j a_\nu \qquad (0 \leq j < \ell).$$

The numbers $\lambda_\nu = \zeta^{\nu s}$ are distinct: for $\nu < \nu'$, $0 < (\nu' - \nu)s \leq (k-1)s < d$. The first k displayed observations recover the a_ν by a Vandermonde inverse, so their stacked map is injective on W. The polynomial $p(z) = \prod_{\nu<k}(z - \lambda_\nu)$ annihilates these sequences wherever all $k+1$ successive observations occur. Taking $m = 2M_2$, the available exponents are exactly $j + 2 \leq mk$.

The monic recurrence $p(R)x = 0$ determines each cyclic channel from its first ks positions: its leading term recovers position n from positions $n - ks, n - (k-1)s, \ldots, n - s$. Thus $\dim \ker p(R) \leq bhks$ on A. This coordinate proof works without diagonalizing the cyclic operator. Apply Lemma 6.5 and $D \geq bdk - r$:

$$m(bdk - r) \leq bhd + (m-1)bhks \leq bM_2kd + (m-1)bM_2k^2 s.$$

Consequently

$$r \geq \frac{bkd}{2} - \frac{b(2M_2 - 1)k^2 s}{2}.$$

Now $(\ell - 1)s \leq N_0 - 1$ and $(2M_2 - 1)k \leq \ell - 1$. The subtracted term is at most $bk(N_0 - 1)/2$, proving Eq. (17).

For each fixed k, once $d \geq \max\{N_0, 2(N_0 - 1)\}$ the bound is $r \geq bkd/4$. Choosing k arbitrarily large before d proves the uniform divergence of $r/(bd)$. □

This excludes repairs of uniformly $O(bd)$ rank, including all odd binary dimensions. It does not give $r \geq \epsilon bd^2$ for any fixed $\epsilon > 0$. The arithmetic-progression cutoff grows with k, and the proof does not permit k to be a fixed positive fraction of d. Nor is rank automatically a physical time charge. Additional stages, nonlinear factors and paid native realization require separate arguments.

7 Complete nonlinear maps: symmetry, balance and periods

The arguments in this section use exactness on the complete finite state space. They do not linearize a decoder on a correctness source. A complete bijection has balanced column outputs; an arbitrary function agreeing with transpose on a subspace need not have them. This distinction supplies an additional obstruction beyond the source-dimension bounds of Part I.

7.1 One nonuniform layer and periodic first maps

Theorem 7.1 (An equivariant sandwich)**.** Lean: `uniform_sandwich_injective_classes`, `uniform_sandwich_finite` [`UniformSandwich.lean`]

Let the entry alphabet have at least two elements. Suppose U, V commute with every column rotation, U is injective, K is class-uniform, and $UKV = \tau$ on the complete state space. Then every receiver class of K is a singleton. If the alphabet is finite, the injectivity of U follows from the factorization and need not be assumed.

Outline of the formal proof. Put $S = \mathrm{Fix}(T)$ and $W = KV(S)$. The set τS consists of matrices with all rows equal, so it is invariant under column rotation. Since $UW = \tau S$, equivariance and injectivity of U imply that W is column-rotation invariant. On each input in $V(S) \subseteq S$, the columns of K agree at any two receivers c, e in the same class. Applying this fact to every rotation of every element of W shows that T^{e-c} fixes all of W. Equivariance of U would then make T^{e-c} fix all of τS. An arbitrary repeated row word disproves this when $c \neq e$. For a finite alphabet, U is surjective because the composite is, and therefore injective. □

The following quantitative class count concerns an actual periodic first map, and permits that map to be nonlinear.

Theorem 7.2 (A periodic first factor)**.** Lean: `covariance_no_class_collision`, `orbit_classes_injective`, `exact_periodic_first_class_bound`, `exact_periodic_first_class_size_bound` [`SubgroupClasses.lean`]

Let A be a finite alphabet with at least two elements and let $\tau = KF$ on $A^{d\times d}$, where F is any total map commuting with T^r and K is class-uniform with M classes. Put $g = \gcd(d, r)$. Every class meets each orbit of addition by r in at most one receiver. Consequently each class has at most g receivers and

$$M \geq d/g. \tag{18}$$

The conclusion includes $r = 0$, when it says only $M \geq 1$.

Outline of the formal proof. Both factors are bijective. From $FT^r = T^r F$ and $KF = \tau$ we obtain $KT^r = R^r K$, hence the same covariance for every $s \in \langle r\rangle$. If $e = c + s$ lies in the same class as c, the shared rule and this covariance give, for every input X,

$$\mathrm{col}_e K(X) = \mathrm{col}_c K(T^s X) = \sigma^s \mathrm{col}_c K(X).$$

Surjectivity of K rules out this fixed relation between two distinct output columns. The g cyclic orbits give the asserted class-size and class-count bounds. The formal proof derives the numerical bound through the order of r in $\mathbb{Z}/d\mathbb{Z}$. □

The bound is sharp for every divisor $g \mid d$. Here is the explicit partial-digit-swap construction, a reuse of the composition examples in [1]. Write $d = gh$, $i = gw + z$ and $c = gu + v$, with $0 \leq z, v < g$ and $0 \leq w, u < h$. Define

$$(FX)_{gw+z,gu+v} = X_{gw+v,gu+z}, \qquad (KY)_{gw+z,gu+v} = Y_{gu+z,gw+v}.$$

Both are coordinate permutations, F commutes with T^g, and direct substitution gives $KF = \tau$. The second map has the h interval classes $p(gu + v) = u$, with

$$f_u(Z)_{gw+z} = Z_{gu+z,\, g(w-u) \bmod d}.$$

Indeed $Z = T^{gu+v}Y$ turns the column index in this formula into $gw + v$. If $\gcd(r, d) = g$, then $\langle r\rangle = \langle g\rangle$, so this also realizes equality for the prescribed subgroup. The construction is algebraic; a coordinate permutation is not a free tape operation.

7.2 Agreement distance from equivariance

The one-layer obstruction has a simple probabilistic form on an arbitrary correctness set. This form uses no linear-source assumption.

Proposition 7.3 (Full-input agreement bound)**.** Lean: `full_input_agreement_probability`, `repeated_class_agreement_probability` [`CompositionAgreement.lean`]

Let A be finite, $|A| \geq 2$, and put $q = |A|^d$, the cardinality of one whole column. If K is class-uniform, with class sizes n_J, then

$$\Pr_X[KX = \tau X] \leq \frac{M}{d} + \frac{\sum_J n_J(n_J - 1)}{2dq}, \tag{19}$$

where X is uniform on the complete state space. If some class has two distinct receivers, the separate bound

$$\Pr_X[KX = \tau X] \leq \frac{1 + q^{-1}}{2} \tag{20}$$

also holds.

Outline of the formal proof. For a fixed X, consider all d rotated inputs $T^s X$ and count the phases on which K is correct. At receiver $c = -s$, its view is X. Correctness there requires

$$f_{p(c)}(X) = z_c(X), \qquad z_c(X) := \sigma^{-c} \operatorname{row}_c X.$$

The z_c are independent uniform words from a set of cardinality q. In one class, the number of correct phases is at most the largest multiplicity of these words, hence at most one plus the number of equal unordered pairs in that class. Taking expectations and using rotation invariance of the uniform input law proves Eq. (19).

For the second bound, take distinct same-class receivers c, e, set $s = e - c$, and let $G = \{X : KX = \tau X\}$. If both X and $T^s X$ lie in G, sharedness gives $\operatorname{row}_e X = \sigma^s \operatorname{row}_c X$. This event has probability $1/q$, since the two rows are independent. Thus $\Pr(G \cap T^{-s} G) \leq 1/q$. Inclusion–exclusion and equality of the two marginal probabilities give $2\Pr(G) - 1 \leq 1/q$. □

For example, let F be any first map, let H be equivariant, and let $\gamma = \Pr[KF \neq \tau]$ and $\epsilon = \Pr[F \neq H]$. The composite KH has the same class partition as K, and is correct on every input where KF is correct and $F = H$. Therefore $\gamma + \epsilon$ is at least one minus either applicable bound above. This measures equality of entire maps on an input. It does not measure the number of output coordinates changed by the first map.

7.3 Balanced intertwiners

We now allow a second nonuniform layer, but retain a concrete property of its first nonuniform operator. Let Σ be a finite alphabet of size $a \geq 2$ with a distinguished symbol 0. For width $b \geq 1$, regard a state as a tuple of db words in Σ^d, one for each row and lane. Let L be a zero-fixing permutation of Σ^d, applied identically to all these words. Define

$$Q = LTL^{-1}, \qquad Q_c = T^c Q T^{-c}.$$

In the next statement the group acts on scalar words; the remaining maps may mix all rows and lanes.

Theorem 7.4 (Transitive pairs obstruct balanced outputs)**.** Lean: `transitive_pair_impossible` [`CompositionWordAction.lean`]

Suppose H is a total column-equivariant map and K is a total class-uniform map on $\Sigma^{d\times d\times b}$, where $d \geq 2$. If $KLH = \tau$ everywhere and $p(c) = p(e)$, the group $\langle Q_c, Q_e\rangle$ cannot act transitively on $\Sigma^d \setminus \{0\}$.

Outline of the formal proof. Exactness on the finite complete domain makes H and K bijective. Writing $q = a^{db}$ for the whole-column alphabet, each column of K has exactly q^{d-1} preimages for each value. Since input rotation is bijective, every shared decoder f_J has these same balanced fibres.

Equivariance of H gives $KQ = RK$. Evaluating at a receiver in J and changing its view yields

$$f_J(Q_c Y) = \sigma f_J(Y) \qquad (c \in J). \tag{21}$$

The corresponding inverse identities also hold. A word in Q_c, Q_e and their inverses therefore rotates f_J by its signed exponent sum. If the word represents the identity in the actual scalar-word group $G = \langle Q_c, Q_e\rangle$, it fixes every matrix. Surjectivity of f_J and the exact order d of σ imply that its exponent sum is zero modulo d. Consequently there is a surjective homomorphism

$$\theta : G \longrightarrow \mathbb{Z}/d\mathbb{Z}, \qquad \theta(Q_c) = \theta(Q_e) = 1.$$

Let $N = \ker\theta$. Equation (21) makes f_J constant on all N-orbits of matrices.

Suppose G were transitive on the nonzero words. Normality of N makes all its orbits there have one size λ, and G/N acts transitively on the set of these orbits. Its cardinality t divides d, so

$$\lambda = (a^d - 1)/t > 1, \qquad \gcd(\lambda, a) = 1.$$

The strict inequality follows from $t \leq d < a^d - 1$ for $a, d \geq 2$. If a matrix Y is nonzero, choose one nonzero row/lane word y. The stabilizers satisfy $N_Y \leq N_y$, hence

$$|N \cdot Y| = [N : N_Y] = [N : N_y][N_y : N_Y] = \lambda[N_y : N_Y].$$

This uses no transitivity on tuples of words. Choose a column value different from $f_J(0)$. Its fibre excludes the zero matrix and is a union of N-orbits, so its size is divisible by λ. But its balanced size q^{d-1} is a power of a, a contradiction. □

Surjectivity alone would give the cyclic quotient but would not give the last divisibility contradiction. Likewise, a code decoder defined on a proper correctness source need not have balanced fibres.

Proposition 7.5 (Anchored binary differences)**.** Lean: `anchored_adjacent_classes_distinct` [`CompositionAnchoredEndpoint.lean`]

For $d \geq 3$, define the binary word permutation

$$(Lx)_0 = x_0, \qquad (Lx)_i = x_i + x_{i+1} \quad (1 \leq i < d),$$

with cyclic indices. In any complete factorization $KLH = \tau$, with H equivariant, the partition of K satisfies $p(c) \neq p(c+1)$ for every receiver c. This holds at every positive width, even when H and K are nonlinear and mix lanes.

Outline of the formal proof. The inverse is $x_0 = y_0$ and $x_i = y_0 + \sum_{j=i}^{d-1} y_j$ for $1 \le i < d$. For $S = \sum_i y_i$, this gives $Qy = (S, y_2, \dots, y_{d-1}, S + y_0)$. Conjugating $\langle Q, TQT^{-1} \rangle$ by L^{-1} gives $\langle T, A \rangle$, where direct substitution in this inverse yields

$$Ax = (x_2, x_1 + x_2, x_0 + x_1, x_3 + \eta, \dots, x_{d-1} + \eta), \qquad \eta = x_0 + x_1 + x_2.$$

On its first triple, A has the cycle

$$100 \to 001 \to 110 \to 010 \to 011 \to 101 \to 111 \to 100.$$

Moreover $(Ax)_j + (Ax)_2 = x_j + x_2$ for every $j \ge 3$. Thus a power taking a nonzero triple to another with the same third bit fixes every outside coordinate, regardless of its value. In particular the following are local moves in the group:

$$100 \to 010, \quad 110 \to 010, \quad 101 \to 001, \quad 011 \to 001, \quad 111 \to 001.$$

They use powers $3, 1, 3, 4, 2$, respectively. Conjugation by rotations places them on any cyclic triple.

For a word with at least two ones, choose consecutive ones in cyclic order. If their intervening zero gap has length at least two, use $100 \to 010$ to shorten it. A gap of length zero or one then permits one of the other moves to lower the weight. Iteration reaches a unit vector, which a rotation sends to coordinate zero. Reversing such paths proves transitivity on all nonzero words, including when $d = 3$. Conjugacy gives the same transitivity for every adjacent pair $\langle Q_c, Q_{c+1} \rangle$. Apply Theorem 7.4. □

7.4 A joint-output period constraint

Balance examines a shared output column. A literal period of the final mask supplies a separate constraint on the complete output map.

Theorem 7.6 (Transport of a mask period)**.** Lean: `literal_period_complete`, `fixed_conjugate_iff`, `fixed_image_preserved` [`PeriodTransport.lean`]

Let the entry alphabet be finite. Suppose U, B, H are total column-equivariant maps, L is any total map, K is class-uniform, and

$$\tau = UKBLH$$

on the complete state space. Exactness makes L bijective. If $p(c + s) = p(c)$ for every c, then its actual conjugated shift $Q = LTL^{-1}$ commutes with T^s.

Outline of the formal proof. All five factors are bijective. Put $F = UKB$. Sharedness and the literal mask period give $KT^s = T^sK$, hence $FT^s = T^sF$. Equivariance of H gives $FQ = RF$. As R commutes with T^s,

$$FQT^s = RFT^s = RT^sF = T^sRF = T^sFQ = FT^sQ.$$

Cancel the injective map F. The formal proof derives the required equivalences and injectivity from the total factorization. □

In particular, T^s must preserve $\mathrm{Fix}(Q) = L\,\mathrm{Fix}(T)$. For the anchored binary map of Proposition 7.5, this is the space of matrices supported on column zero: $L(1, \dots, 1) = e_0$ on each row/lane word. No nonzero column rotation preserves this space.

Corollary 7.7 (The anchored two-class architecture)**.** Lean: `anchored_at_most_two_classes_impossible` [`CompositionAnchoredEndpoint.lean`]

For every $d \geq 3$ and every positive binary width, there is no complete factorization $\tau = KBLH$ in which L is the anchored difference, B, H are arbitrary equivariant total maps, and K uses at most two receiver classes.

Outline of the formal proof. Absorb B into K; this preserves its shared partition. By Proposition 7.5, that partition is a proper coloring of the cycle. An odd cycle has no proper two-coloring. On an even cycle, every proper two-coloring alternates and has period two. This is a nonzero period for even $d \geq 4$, contradicting Theorem 7.6 and the preceding fixed-space calculation. □

The period theorem permits an arbitrary equivariant outer U. The combined anchored corollary does not: its balance step concerns K itself. Neither result excludes two arbitrary nonlinear nonuniform layers, gives a positive agreement-rate deficit for the anchored architecture, or supplies a native running-time bound.

8 Construction requirements and the remaining questions

The results constrain two distinct resources: one-stage correctness dimension and complete-state composition. Their combination removes several ways of turning a large one-stage source into a complete algorithm, but it does not supply an interface theorem for arbitrary native tape computations.

For [1, Problem 9.1(a)], the next capacity question is whether every fixed number of shared deterministic decoders has dimension at most $N/2 + o(N)$ on linear sources. The two-class 41/64 bound leaves a constant gap; the fixed-class constructions give only vanishing normalized excess. Nonlinear source promises and chosen-source nonlinear-to-linear losses must not be substituted for this maximum-capacity question. The measured covariance and symmetric-retention deficits provide more structured intermediate targets. The exact arbitrary-set capacity shows why a high-entropy fibre alone is insufficient: a native extraction needs the linearity or actual affine-intersection premise used by its capacity bound. The mask floors identify limits of the stated positive-term certificates; improving beyond them requires more compatibility information, a smaller actual starting ambient, or another argument.

The interval constructions attain the bound with deterministic binary decoders when $d - M$ is even. This does not settle the original linear-decoder attainment question for unequal pieces in [1, Problem 9.2]. The diagonal-defect bound supplies a way to retain actual block losses when testing that question.

For composition, a sharp class-product inequality for arbitrary linear masks would improve the Ramsey cutoff. A positive normalized rank deficit would strengthen the present growing rank loss. On the nonlinear side, the complete masked-neighbor family remains open beyond the uniform, anchored and single-site cases and the general period restrictions. The balance arguments apply to actual conjugated operators and balanced fibres; an assigned covariance equation or a surjective but unbalanced decoder is insufficient.

The complete scattered first factor is a concrete warning: total bijective completion need not preserve a small class count in the required second map. The residual bounds offer a different application route, if one can exhibit a complete structured affine baseline and pay its joint correction record. They do not supply such a baseline for an arbitrary fast computation.

Finally a native application must account for production of the representation, access to its entries, the complete state at every intermediate stage, and recovery of the ordered output. None of

the shared-rule theorems proves that a linear-time native transposer would possess the required fixed class count, linear source or same-space factorization. This is the unresolved step toward [1, Problem 9.4]. A native theorem giving one of those interfaces with small paid advice could make the algebraic bounds decisive. A construction must instead realize a complete compatible sequence of rules within the native time budget. The two-stage theorems do not exclude a growing number of stages, and none of the bounds supplies the multiscale time charge asked for in [1, Problem 9.5].

A Exact capacity for arbitrary correctness sets

The linear-source premise in the capacity theorems cannot be replaced by a large correctness set. Here the alphabet Σ need only be finite and nonempty, with q symbols. Let $d \geq 1$, let σ rotate columns as in Section 2, and write x_c for row c of X. A single total deterministic decoder f validates a set C when

$$f(\sigma^c X) = x_c \qquad (X \in C,\ c \in \mathbb{Z}/d\mathbb{Z}).$$

Let $M(d, q)$ be the largest possible $|C|$. There is no linearity or evaluation-cost restriction on f.

For each matrix define subgroups

$$H_X = \{h : \sigma^h X = X\}, \qquad K_X = \{a : x_{c+a} = \sigma^a x_c \text{ for every } c\}.$$

The second rotation acts on a row word. Call the column-rotation orbit of X eligible if $H_X \subseteq K_X$. Both groups are unchanged when X is replaced by $\sigma^t X$.

Theorem A.1 (Exact one-class set capacity)**.** Lean: `SetCapacity.exact_capacity`, `SetCapacity.capacity_orbit_formula`, `SetCapacity.capacity_average_formula` [`CapacitySetOrbit.lean`]

With the preceding conventions,

$$M(d, q) = \sum_{\text{eligible orbits } O} \frac{|K_O|}{|H_O|} = \frac{1}{d} \sum_X \mathbf{1}_{H_X \subseteq K_X} |K_X|. \tag{22}$$

One total decoder attains the bound simultaneously on all orbits.

Outline of the formal proof. H_X is a stabilizer. The covariance defining K_X is preserved under addition and negation of shifts, so K_X is also a subgroup. Column rotations commute, proving invariance of both groups along an orbit.

A single matrix X is admissible exactly when repeated views prescribe the same output. Repetition at rows $c, c + h$ means $h \in H_X$, and then the two demanded rows must agree. Since such h fixes every row word, this is exactly $H_X \subseteq K_X$.

In an eligible orbit write $Y_a = \sigma^a X$, with a taken modulo H_X. Two orbit words Y_a, Y_b are compatible exactly when $a - b \in K_X$. Indeed their views at $c, c' = c + a - b$ coincide. Equality of the required outputs is $\sigma^a x_c = \sigma^b x_{c+a-b}$, precisely the covariance for $a - b$. Conversely, any equality of views has $c' = c + a - b + h$ with $h \in H_X$; membership of $a - b, h$ in K_X makes the outputs agree. Thus all collisions, including collisions within one word, are accounted for.

A nonempty compatible subset of the orbit lies in one coset of K_X/H_X, and the whole coset is compatible. Its maximum size is $|K_X|/|H_X|$. Choose one such coset in every eligible orbit. Views from different orbits cannot collide, so all their output requirements define one partial function. Extend it arbitrarily to a total decoder. This proves the first formula and attainment. Each orbit has $d/|H_X|$ elements; summing the constant value $|K_X|$ on it gives the second formula. For $d = 1$ the value is q, and for $q = 1$ it is 1, as the same formula asserts. □

The formal construction chooses an actual representative in every rotation orbit, takes its entire covariance coset when eligible, and proves all its view requirements consistent before totalization. Its cardinal and the universal upper bound are both obtained from actual finite fibres. The proof uses consistency of deterministic function values. It supplies neither the additional additive consistency needed for a linear decoder nor a small description or cheap evaluator for the attaining function.

Corollary A.2 (Prime dimensions and the all-dimension rate)**.** Lean: `SetCapacity.prime_capacity`, `SetCapacity.fixed_H_card`, `SetCapacity.fixed_K_card` [`CapacitySetCounting.lean`]
Lean: `SetCapacity.relative_error`, `SetCapacity.capacity_lower`, `SetCapacity.logarithmic_rate_tendsto` [`CapacitySetRate.lean`]

For prime d,

$$M(d,q) = \frac{q^{d^2} + (d-2)q^d + q}{d}.$$

For all $d \geq 2, q \geq 2$,

$$\left| \frac{dM(d,q)}{q^{d^2}} - 1 \right| \leq (d-1)q^{-d\lceil d/2 \rceil}, \qquad M(d,q) \geq \frac{q^{d^2}}{2d}.$$

Consequently, at fixed $q \geq 2$ along all integer dimensions, $\log_q M(d,q) = d^2 - \log_q d + o(1)$.

Outline of the formal proof. For prime d, each subgroup is either trivial or the whole group. There are q^d matrices with full H (constant rows), q^d with full K (choose x_0 and set $x_c = \sigma^c x_0$), and q with both full. The four cases (H, K) therefore contribute: q words when both are full; $q^d - q$ when only K is full; zero when only H is full; and $(q^{d^2} - 2q^d + q)/d$ when both are trivial. Their sum is the formula.

For a nonzero shift h, put $g = \gcd(d, h)$. The condition $h \in H_X$ permits g free symbols in each row, hence q^{dg} matrices. The condition $h \in K_X$ has the same count: choose one whole row in each of the g row cycles and propagate by rotation. On returning around a cycle the total rotation is a multiple of d, so there is no extra constraint. Put $S = \sum_{h \neq 0} q^{d \gcd(d,h)}$. A union bound leaves at least $q^{d^2} - S$ matrices with trivial H; each contributes at least one to the second sum in Eq. (22). Dropping eligibility and summing $|K_X|$ gives the upper bound $dM(d,q) \leq q^{d^2} + S$. Since $\gcd(d,h) \leq \lfloor d/2 \rfloor$, $S \leq (d-1)q^{d\lfloor d/2 \rfloor}$. Division gives the displayed relative error. It is at most $1/2$ and is bounded by $d/2^d$, which tends to zero. The lower bound follows, and continuity of $\log_q$ at 1 gives the logarithmic remainder. The formal limit runs through dimensions $n+1$, so it covers every positive integer dimension. No extrapolation from small dimensions is used. □

A.1 Line views, linear sources and affine repairs

For a line, rows and columns are $0, \ldots, d-1$ and the full view is

$$\mathrm{view}_c(X)_{i,e} = \begin{cases} X_{i,e+c} & 0 \leq e + c < d, \\ 0 & \text{otherwise}, \end{cases} \qquad -d < e < d.$$

Fix a distinguished alphabet symbol 0. The Lean line view uses every integer offset; for a legal receiver all offsets outside the displayed window are identically zero. Restriction and zero extension therefore give the same decoder interface.

Proposition A.3 (High-density line marker)**.** Lean: `LineMarker.exists_decoder`, `LineMarker.code_card`, `LineMarker.affineMarker_finrank` [`CapacityLineMarker.lean`]
Lean: `LineMarker.code_density`, `LineMarker.code_density_tendsto` [`CapacityLineMarkerRate.lean`]

One total deterministic line decoder works on $C = \{X : \text{the first column is nonzero}\} \cup \{0\}$*, whose size is*

$$|C| = (q^d - 1)q^{d(d-1)} + 1.$$

Over a field, its affine subset $X_{0,0} = 1$ *has dimension* $d^2 - 1$*. No optimality claim for this line construction is made.*

Outline of the formal proof. In the view of a matrix with nonzero first column, the leftmost offset with any nonzero symbol is exactly $-c$. If two such views agree, applying this boundary test in both directions proves that their receivers agree; evaluating at offsets $j-c$ then proves their matrices agree. A view equal to the all-zero view comes only from the zero matrix. Thus all required outputs are consistent, and one total decoder follows. A nonzero first column has q^d-1 choices; all other columns are free. Fixing one entry to 1 leaves an affine space of the asserted dimension. Its dimension exceeds the one-class half bound for $d \geq 3$, as checked by `LineMarker.affineMarker_exceeds_half`; the arbitrary set has density $1 - q^{-d} + q^{-d^2}$, tending to one. □

Theorem A.4 (Deterministic line rules on linear sources)**.** Lean: `IsDeterministicLineCode.two_mul_finrank_le`, `IsDeterministicLineCode.affine_two_mul_finrank_le` [`CapacityLineDeterministic.lean`]

Let $C \subseteq F^{d\times d}$ *be a linear subspace of dimension* D*. Suppose fixed deterministic decoders, constant on each of* M *nonempty consecutive row intervals, recover every demanded row from its line view. The labels may repeat on different intervals. Then*

$$2D \leq d^2 + Md.$$

Outline of the formal proof. Let T shift columns left with a zero final column, and let Z be the subspace with zero first column. Set $I = C \cap T^{-1}C \cap Z$. If $\operatorname{rank} T = r$, then $\dim T^{-1}C = d^2 - r + \dim(C \cap \operatorname{im} T) \geq D$. Hence $\dim I \geq 2D - d^2 - d$. For $X \in I$ the zero first column gives the exact view identity $\operatorname{view}_c(TX) = \operatorname{view}_{c+1}(X)$. Within one interval, identical arguments to the same decoder imply $x_{c+1} = Tx_c$. In an interval of length ℓ, all rows are shifts of its first row, and their zero first entries force its first ℓ entries to vanish. At most $d - \ell$ entries remain free. Thus $\dim I \leq \sum(d - \ell) = Md - d$, giving the bound. Only equality of decoder values, not their linearity, was used. □

This extends the linear-decoder line proof of [1, theorem "The line form of Theorem TP"]; the source remains linear. On a cycle an affine source $x_0 + V$ obeys the corresponding translate intersection bound whenever the *joint* intersection of the selected rotated sources is nonempty. Choose a point of that intersection; its direction is the intersection of the rotated copies of V, and the same codimension and homogeneous covariance bounds apply. Pairwise intersection does not suffice for three or more shifts. The affine line counterpart is also formalized by `IsDeterministicLineCode.affine_two_mul_finrank_le`. Its sufficient affine premise is instead the nonempty threefold slice $C \cap T^{-1}C \cap Z$, because the view identity needs the boundary zero. The marker $X_{0,0} = 1$ has $C \cap T^{-1}C \neq \varnothing$ for $d \geq 2$ but cannot meet Z, demonstrating the missing premise.

If the decoders themselves are affine, subtracting their equations at $x_0 + v$ and x_0 produces a linear shared rule on V. That separate argument needs no affine-intersection premise; it is invalid

for arbitrary deterministic decoders. The revised affine-entry advice theorem in [1, Corollary 6.16] uses precisely such cancellation. Our arbitrary-set constructions therefore do not contradict it.

These results sharpen the model boundary relevant to native extraction: high entropy alone does not provide a linear source or the required affine intersection. Conversely, recognizing the receiver from an entire static view does not provide a paid native operation that reads that view.

B Diagonal defects and exact displayed kernels

The block flag bound can retain the actual defect on each diagonal block. This is useful when a proposed column algebra has small invariant blocks, but its diagonal compressions admit fewer transpose agreements than their full dimensions would allow.

Theorem B.1 (Unequal block defects)**.** `Lean: upperBlockDescriptor_injective, finrank_upperBlockAgreement_le` [`ConstructionBlockDefects.lean`]

Let $V = \bigoplus_{j=1}^{m} V_j$, with $\dim V_j = s_j$, and use compatible bases on the two tensor factors. In these coordinates suppose the linear operator K has block coefficients

$$(KX)_{ij} = \sum_{a,b} L_{ab}^{ij}(X_{ab}), \qquad L_{ab}^{ij} = 0 \quad \text{when } b < j.$$

Write $K_j = L_{jj}^{jj}$ for its actual diagonal compression and put $\delta_j = \dim\ker(\tau_{s_j} - K_j)$. Then

$$\dim\ker(\tau - K) \le \sum_{i<j} s_i s_j + \sum_j \delta_j.$$

In particular this applies when the column factors preserve the prefix flag: that hypothesis gives exactly the displayed coefficient vanishing. For b coupled lanes the first summand becomes $b\sum_{i<j} s_i s_j$; each δ_j is the full diagonal-block defect including its lanes.

Outline of the formal proof. Retain the blocks strictly above the diagonal. On each diagonal block retain a linear projection onto the kernel of its actual diagonal compression. We show that this descriptor is injective on the agreement space. Subtract two inputs with the same descriptor. Their upper blocks vanish. Descend through the remaining block rows. The flag property eliminates earlier columns; every other term is already zero by the upper-block condition or the induction. The diagonal equation therefore puts the current diagonal block in its diagonal kernel. Its retained projection is zero, so the block itself is zero. The remaining equations then set the entire current block row to zero. The descriptor has the stated dimension.

The Lean statement allows different vector spaces on each rectangle, explicit transpose equivalences, and all coefficients from later block columns. It thus includes the tensor formulation and upper-triangular coupling. The weaker block flag bound follows by replacing each δ_j by s_j^2 (or bs_j^2). □

B.1 The full kernel of the two-class cyclic lift

The exhibited seed and lift also admit an exactness statement, stronger than the existence of a subcode of the same dimension. It concerns these particular operators; it does not optimize over seed decoders.

Theorem B.2 (Exact displayed cyclic-lift kernel)**.** Lean: `binarySeed_full_rule`, `mem_cyclicLiftCode_iff_full_rule`, `binaryCyclicDefect_ker`, `finrank_binaryCyclicLaneDefect_ker` [`ConstructionCyclicExact.lean`]

For the published binary 6×6*, two-class seed, the full correctness set of its displayed linear decoders is its* 25*-dimensional code. For* $d = 6m$*,* $\gcd(m, 6) = 1$*, and every width* b*, the entire agreement kernel of the displayed cyclic lift, applied independently to the* b *lanes, has dimension*

$$b(18m^2 + 7m) = b\left(\frac{d^2}{2} + \frac{7d}{6}\right).$$

More generally, the cyclic lifting construction preserves equality between a seed code and the full correctness set of its deterministic decoders, whenever its averaging scalar is invertible. This generic statement is `mem_liftedCode_iff_full_rule`*; the cyclic-index counterpart cited above uses linear seed decoders.*

Outline of the formal proof. Let A_6 be the defect map whose (c, r) coordinate is the demanded seed entry minus the displayed decoder entry. The published generators already provide a 25-dimensional subspace of $\ker A_6$. Their private pivot coordinates define a linear expansion P into that subspace. A sparse linear certificate R satisfies

$$P + RA_6 = I.$$

Lean checks this identity on the 36 standard basis matrices using ordinary kernel evaluation. Hence $A_6X = 0$ implies $X = PX$ and supplies the reverse inclusion. The certificate generator does not belong to the trusted proof; its output is checked again by Lean. The declarations `binarySeed_recovery_identity` and `finrank_binarySeedDefect_ker` record this step.

For the lift, let E be the anti-diagonal averaging projector used in Section 5. The displayed rule writes a correct input as

$$X = \tau(I - E)X + \mathrm{inflate}(R(\mathrm{average}\, X)),$$

where R applies the displayed seed rules to every averaged slice. Average this identity. The first term averages to zero, and averaging undoes inflation. Every averaged slice therefore satisfies all seed rules and belongs to the seed code. Subtract its inflation: $(I - E)X$ is symmetric and has zero average. These are precisely the two summands of the constructed lifted code. The converse is the existing decoder correctness proof. The Chinese remainder equivalence transfers both directions to literal cyclic indices. The previous dimension theorem therefore computes the full kernel, and independent lanes multiply its dimension by b. □

For example, at $d = 30, b = 1$ the full kernel has dimension 485, compared with the two-interval ceiling 480 and the identity kernel dimension 465. These are consequences of the general theorem, not separate numerical evidence for it.

Reproduce the finite certificate with `python3 tools/seed_recovery_certificate.py` from this paper's directory, then run `lake build` from `lean/`. The seed data are read from the sibling Paper 17 tools directory. The paid native schedule in Theorem 5.4 is unchanged: exact kernel equality adds no new operation or hidden decoder oracle.

C Mask-sensitive bounds and limits of the certificate methods

Throughout this section C is a linear source of dimension D over an arbitrary field, entries have dimension b, and the fixed shared decoders may be nonlinear. Write $N = bd^2$. Probabilities sample

receiver labels, never inputs to a machine or native algorithms. A *certificate* below is the numerical right side of a specified upper-bound inequality. Lower bounds on these numbers are not lower bounds on achievable code dimension.

C.1 Finite and structured masks

For a binary mask p let $R_r(s)$ count starts of monochromatic runs of length at least r along addition by s: a start c satisfies $p(c-s) \neq p(c) = p(c+s) = \cdots = p(c+(r-1)s)$. Constant translation orbits have no such starts. Put $Q_s = R_1(s)$, the number of directed color changes.

Proposition C.1 (A finite alternative)**.** Lean: `finite_weighted_run_budget`, `IsSharedCode.four_mul_finrank_le`, `IsSharedCode.six_mul_finrank_le` [`CapacityMaskFinite.lean`]

For a binary partition and $d > 8$,

$$6D \leq 4bd^2 + 5bd.$$

Outline of the formal proof. The two-stage nested inequality with shifts $(1,3)$ gives $4D \leq 2N + bd(R_1 + R_4 + 3)$. With shifts $(2,1)$ it gives $4D \leq 2N + bd(2R_2 + R_3 + 4)$: use $Q_2 = 2R_2$, and retain the possible two translation orbits of the step 2. The run counts here have step 1. Two copies of the first inequality plus one of the second use the budget

$$2R_1 + 2R_2 + R_3 + 2R_4 \leq 2d.$$

For completeness, charge the two length-one tickets of a run to offset 0 in lanes $0, 1$, its two length-two tickets to offset 1 in those lanes, its length-three ticket to offset 2 in lane 0, and its two length-four tickets to offsets $2, 3$ in lanes $1, 0$. Tickets occur only when the charged offset is within that run. Distinct runs cannot collide, and tickets within one run have distinct position–lane pairs. Thus they inject into two copies of the cycle. Substitution gives $12D \leq 8N + 10bd$. □

Theorem C.2 (An explicit family of masks)**.** Lean: `CapacityMaskStructured.first_component_count`, `CapacityMaskStructured.second_component_count`, `CapacityMaskStructured.structured_finrank_le` [`CapacityMaskStructured.lean`]

Let $d = 8\ell$, $\ell \geq 1$, write $c = a + \ell r$ with $0 \leq a < \ell$ and $r \in \mathbb{Z}/8$, and let

$$p(c) = v(a) \operatorname{xor} u(r), \qquad u = (0,1,1,0,1,0,0,1),$$

where v is arbitrary. Then

$$D \leq \frac{9}{16}bd^2 + \frac{1}{2}bd.$$

Outline of the formal proof. Use nested shifts 3ℓ and 1. The first graph has exactly 2ℓ components. For each fixed a, its cyclic order of high indices is $0, 3, 6, 1, 4, 7, 2, 5$, whose labels are $0, 0, 0, 1, 1, 1, 1, 0$ up to a common complement. Each color therefore induces one four-vertex path.

The second shift set is $\{0, 3\ell, 1, 3\ell + 1\}$. Its differences retain the first graph and include $3\ell + 1$. For $a < \ell - 1$, that step sends (a, r) to $(a + 1, r + 3)$. For every $z, \delta \in \{0, 1\}$ there is an r with $u(r) = z$ and $u(r + 3) = z \operatorname{xor} \delta$: use $r = 0, 6, 1, 2$ for $(z, \delta) = (0,0), (0,1), (1,0), (1,1)$, respectively. Taking z to be the desired global color xor $v(a)$ and $\delta = v(a) \operatorname{xor} v(a + 1)$ connects that color's two neighboring low-index components. All low indices are thereby connected within each color. No edge joins colors, so the final graph has exactly two components. When $\ell = 1$ this was already true of the first graph.

The nested bound is $D \le N/2 + n_1/4 + n_2/4$, with $n_1 = bd^2/4$ and $n_2 = 2bd$. Substitution proves the claim. In particular, binary digit parity at $d = 2^h$, $h \ge 3$, belongs to this family: u is the parity of the three high bits and v the parity of the remaining bits. The formal proof constructs classifiers of the actual graph quotients, using the displayed paths and bridges to prove their fibers are exactly the connected components. □

Proposition C.3 (Exact direct envelope for digit parity)**.** Lean: `CapacityDigitParity.digit_parity_envelope`, `CapacityDigitParity.digit_parity_floor_envelope` [`CapacityDigitParityEnvelope.lean`]

For cyclic binary digit parity at $d = 2^h$, $h \ge 3$, the minimum over all nonempty shift sets Δ of the original direct translate certificate is exactly $5bd^2/8$. This is the numerical envelope of that formula, not the capacity of the mask.

Outline of the formal proof. Put $s_h(c) = (-1)^{p(c)}$ and $\eta_h(k) = d^{-1}\sum_c s_h(c)s_h(c+k)$. The formal mask is the literal parity of the lowest h binary digits, defined by repeated division by two. Its periodicity identifies the finite correlation with the cyclic mask statistic. Splitting the average into even and odd positions gives

$$\eta_h(2k) = \eta_{h-1}(k), \qquad \eta_h(2k+1) = -\tfrac{1}{2}(\eta_{h-1}(k) + \eta_{h-1}(k+1)).$$

Every correlation lies in $[-1, 1]$, since its summands are signs. The recurrences show that each adjacent sum at level $h \ge 1$ is either $(\eta_{h-1}(k) - \eta_{h-1}(k+1))/2$ or its negative. It therefore lies in $[-1, 1]$. Every odd-index correlation at level $h \ge 2$ lies in $[-1/2, 1/2]$. Even indices reduce to the preceding level, and the nonzero correlations at $h = 2$ are $0, -1, 0$. Induction gives $\eta_h(t) \le 1/2$ for every nonzero cyclic shift. Consequently

$$Q_t = \tfrac{d}{2}(1 - \eta_h(t)) \ge d/4.$$

The actual first-graph inequality $\chi(\{0, t\}) \ge Q_t$ suffices for the lower bound. Translating a shift set leaves its link relation unchanged, so every two-element set has certificate at least $5N/8$. For $q = |\Delta| \ge 3$, the ambient term alone gives $2N/3$; a singleton set gives N.

For attainment, set $\ell = d/8$ and split the three high bits from the remaining bits. The literal parity mask is exactly the preceding structured family, with u the three-bit parity word. Its actual first graph for $t = 3\ell$ has $2\ell = d/4$ components. This two-element shift set therefore attains $5N/8$. The value is integral, so flooring any other certificate cannot improve it. The structured nested bound improves this entire envelope by $bd(d-8)/16$ when $d > 8$. □

C.2 Run spectra and random masks

We use the bounded-differences inequality in its following precise form. For independent coordinates, changing coordinate i changes a real function f by at most c_i; then either tail at distance $t \ge 0$ from its mean has probability at most $\exp(-2t^2/\sum_i c_i^2)$ when $\sum_i c_i^2 > 0$. This is McDiarmid's inequality[5]; the displayed form and the lower-tail reduction by $-f$ are stated in [8, Theorem 1, equation (2)]. No independence between the individual summands of f is required. For the finite uniform product laws used here, the shared `lean-machine` modules `FiniteUniformProbability`, `FiniteBoundedDifferences`, and `FiniteUniformWindows` prove the needed inequality and restriction identities. The exponential-moment proof applies the bounded-interval estimate successively to actual finite conditional averages; it retains the full interval length c_i at each coordinate.

Proposition C.4 (The three-stage spectrum)**.** Lean: `IsSharedCode.run_spectrum_three` [`CapacityMaskRunSpectrum.lean`]

For $d > 1$, putting $R_r = R_r(1)$ gives

$$D \le \frac{N}{2} + bd\left(\frac{R_1}{4} + \frac{R_4}{8} + \frac{R_{13}}{8}\right) + \frac{3bd}{4}.$$

Outline of the formal proof. Use the actual shifts $1, 3, 9$ in the three-stage inequality. Their successive difference sets cover radii $1, 4, 13$. The corresponding component counts are at most $R_1 + 1, R_4 + 2, R_{13} + 2$, respectively; the first step has a single translation orbit. Their weights are $1/4, 1/8, 1/8$, so the additive error is $3bd/4$. □

Theorem C.5 (Almost every binary mask)**.** Lean: `CapacityMaskProbability.spectrum_bad_probability`, `CapacityMaskProbability.SpectrumGood.source_bound` [`CapacityMaskRunProbability.lean`]

Let $d > 13$, $\varepsilon \ge 0$, and sample the d mask bits independently and fairly. Except on probability at most $\exp(-128\varepsilon^2 d/529)$, simultaneously every valid source and decoder family for the sampled mask satisfy

$$D \le \left(\frac{41473}{65536} + \varepsilon\right) bd^2 + \frac{3bd}{4}.$$

Outline of the formal proof. A run-start indicator of length $r < d$ examines $r + 1$ distinct bits, with exactly two favorable assignments. Thus $\mathbb{E}R_r = d/2^r$. Changing one bit affects at most $r + 1$ of these indicators. For $F = R_1/4 + R_4/8 + R_{13}/8$, the coordinate-change constant is therefore $2/4 + 5/8 + 14/8 = 23/8$, while $1/2 + \mathbb{E}F/d = 41473/65536$. Apply bounded differences with $t = \varepsilon d$ and then the preceding deterministic inequality. The event depends only on the mask, so there is no union over fields, widths, codes or decoders. Conditioning on both colors occurring divides the failure bound by $1 - 2^{1-d}$; identifying complementary labelings does not change any statistic. □

The same reasoning gives the full fixed-depth spectrum. For $k \ge 2$, $d > (3^k - 1)/2$, set $r_j = (3^j - 1)/2$ and use shifts 3^{j-1}. Balanced-ternary digits show by induction that their difference set covers every integer from $-r_j$ to r_j: the preceding interval and its translates by $\pm 3^{j-1}$ are consecutive and exhaustive. Each difference digit is the difference of two digits in $\{0, 1\}$. For a color in cyclic order, a gap greater than r corresponds to an intervening opposite-color run of length at least r; its components number the larger of one and its number of long gaps. Summing both colors gives $\chi_r \le R_r + 2$; at radius one the sharper $R_1 + 1$ also covers constant masks. Consequently

$$D \le \frac{N}{2} + bd\left(\sum_{j=1}^{k-1} \frac{R_{r_j}}{2^{j+1}} + \frac{R_{r_k}}{2^k}\right) + \frac{3bd}{4}.$$

Replace each run count by its mean to obtain the constant $c_k = 1/2 + \sum_{j=1}^{k-1} 2^{-r_j - j - 1} + 2^{-r_k - k}$. The coordinate-change constant is $L_k = \sum_{j=1}^{k-1}(r_j + 1)/2^{j+1} + (r_k + 1)/2^k$, so for $\varepsilon \ge 0$ the upper tail at εd is at most $\exp(-2\varepsilon^2 d/L_k^2)$. The formal endpoints are `CapacityTernarySpectrum.Good.source_bound` for the source inequality and `CapacityTernarySpectrum.bad_probability` for its probability bound. Their proofs compute graph counts, means and sensitivities on the actual recursively generated translate sets.

C.3 What the numerical methods cannot certify

For nonnegative ambient sizes n_j define the exact nested certificate

$$B_k(n) = \sum_{j=0}^{k-1} \frac{n_j}{2^{j+1}} + \frac{n_k}{2^k}, \qquad B_0(n) = n_0.$$

If $n_1 \le n_0$, every $k \ge 0$ satisfies $B_k(n) \ge n_0/2 + n_1/4$. For $k \ge 2$ these are the first two terms; for $k = 1$ the first ambient has weight $1/2$; for $k = 0$ use $n_1 \le n_0$.

Theorem C.6 (The nested-component floor)**.** `Lean`:
`CapacityMaskProbability.component_floor_probability`,
`CapacityMaskProbability.ChangeGood.certificate_floor`
[`CapacityMaskComparisonProbability.lean`]

For $\eta > 0$, outside probability $(d-1)\exp(-\eta^2 d/2)$ under independent fair binary labels, every nested component certificate starting from $n_0 = N$ is at least $(5/8 - \eta/4)N$. Shifts and chain lengths may depend on the whole mask. Taking $\eta = \sqrt{6 \log d/d}$ gives failure probability less than d^{-2} for $d > 1$.

Outline of the formal proof. For every nonzero s, $\mathbb{E}Q_s = d/2$. A bit change affects at most the two directed comparisons at that bit and its predecessor, so the coordinate-change constant is 2, even for a step of order two. Bounded differences and a union over the $d-1$ nonzero steps give one event on which all $Q_s \ge (1/2 - \eta)d$. Every first graph has $\chi(\{0, s\}) \ge Q_s$: on a nonconstant translation cycle its monochromatic components equal its directed changes; a constant cycle adds a component and no change. The formal proof obtains the same inequality by counting the actual retained step edges. Thus $n_1 \ge (1/2 - \eta)N$; for a zero shift $n_1 = N$ instead. Apply the first-term bound for every chain. Finally $(d-1)e^{-\eta^2 d/2} = (d-1)d^{-3} < d^{-2}$ for the stated η. □

Theorem C.7 (The full-ambient isolation floor)**.** `Lean`:
`CapacityMaskProbability.isolation_rate_probability`,
`CapacityMaskProbability.IsolationGood.rounded_floor` [`CapacityMaskMethodEndpoints.lean`]

For $\eta \ge 0$, start the isolation recurrence from all $d-2$ *nonrepresentative rows of a nonempty binary partition. Write n_j for the number of these rows isolated in the graph of the jth nested shift set, in row units, and use precisely $B_k(n)$ with this terminal ambient. Outside probability $(d-1)\exp(-2\eta^2 d/9)$ under independent fair labels, every resulting completed numerical bound $D_{\mathrm{cert}} = bd(2 + B_k(n))$ satisfies*

$$\frac{D_{\mathrm{cert}}}{N} \ge \frac{9}{16} - \frac{\eta}{4} + \frac{1}{2d}.$$

Final integer flooring preserves the safe bound $9/16 - \eta/4 - 1/(2d)$ for every $b \ge 1$. With $\eta = \sqrt{(27/2)\log d/d}$ the unconditional failure probability is less than d^{-2}. Conditioning on both classes being nonempty divides it by $1 - 2^{1-d}$.

Outline of the formal proof. For $s \ne 0$, a vertex is isolated in $G_{\{0,s\}}$ precisely when its label differs from both neighbors at $\pm s$. If s has order two the neighbors coincide and its probability is $1/2$; otherwise three distinct fair bits give probability $1/4$. Hence the total I_s has mean at least $d/4$. A bit change affects at most the indicators centered at that bit and its two neighbors, so its coordinate-change constant is 3. Bounded differences and a union over shifts give, simultaneously, $I_s \ge d/4 - \eta d$. Deleting any two representatives loses at most two isolated vertices. Thus $n_1 \ge d/4 - \eta d - 2$, independently of how the representatives were selected. The zero shift has $n_1 = d - 2$ and also satisfies this inequality. Since $n_0 = d - 2$ and all later sizes are nonnegative, the first-term bound

gives $B_k \geq 9d/16 - \eta d/4 - 3/2$. Add the baseline 2, multiply by bd, and divide by N. Final integer flooring of D_{cert} reduces its normalized value by less than $1/N \leq 1/d$. The stated logarithmic choice gives $(d-1)e^{-2\eta^2 d/9} = (d-1)d^{-3} < d^{-2}$. □

Both floors are preserved under minima, infima over nonempty families, convex averages, and pointwise limits of these same certificates. Replacing counts by upper estimates only raises their right sides. Neither theorem covers a smaller actual starting support, a sharper terminal estimate, additional compatibility equations, or signed combinations of inequalities.

Theorem C.8 (The direct method with many classes)**.** `Lean`: `CapacityMaskProbability.direct_floor_probability`, `CapacityMaskProbability.EqualityGood.direct_gain_inverse_sqrt` [`CapacityMaskComparisonProbability.lean`]

Let $d \geq 2$, $1 \leq M \leq d$, and give the d positions independent uniform labels in $\{1, \dots, M\}$. For a nonempty shift set Δ, put

$$g_\Delta = \frac{d - \chi(\Delta)}{|\Delta| d}, \qquad g_{\mathrm{direct}} = \max_{\varnothing \neq \Delta \subseteq \mathbb{Z}/d} g_\Delta.$$

For $\eta > 0$, except on probability at most $(d-1)e^{-\eta^2 d/2}$,

$$g_{\mathrm{direct}} \leq \sqrt{\frac{1/M + \eta}{2}}, \qquad \min_\Delta N(1 - g_\Delta) \geq N\left(1 - \sqrt{\frac{1/M + \eta}{2}}\right).$$

In particular, for $\eta_d = \sqrt{6 \log d / d}$, the failure probability is less than d^{-2}. If $M\eta_d \leq 1$, then $g_{\mathrm{direct}} \leq 1/\sqrt{M}$.

Outline of the formal proof. For each nonzero s, let $A_s = |\{c : p(c+s) = p(c)\}|$. The endpoints are distinct, so $\mathbb{E}A_s = d/M$. A coordinate change affects at most two comparisons, even for an order-two step. Bounded differences and a union over nonzero steps give one event on which $A_s \leq d(1/M + \eta)$ for all of them.

Fix any mask on this event, then any shift set of size q. Choose one orientation for each of its $\binom{q}{2}$ unordered distinct pairs. All graph edges generated by that pair number at most A_s for the chosen difference s. An order-two step double-counts an edge and repeated differences may count it again; both only increase this upper bound. Every graph component admits a spanning tree, so

$$d - \chi(\Delta) \leq |E(G_\Delta)| \leq \binom{q}{2} d(1/M + \eta).$$

The formal proof gives the same deficit inequality by the rank of the edge-incidence map: its kernel is exactly the functions constant on the actual translate-link quotient. Put $a = 1/M + \eta$. Dividing and also using $d - \chi \leq d$ gives $g_\Delta \leq \min((q-1)a/2, 1/q) \leq \sqrt{a/2}$. The last step follows from $g_\Delta^2 \leq (qa/2)g_\Delta \leq a/2$; the gain is nonnegative. This controls all sets from one event, even when chosen after inspecting the labels. Substitute η_d, and use $a \leq 2/M$ when $M\eta_d \leq 1$. □

An absent label must not be counted as a nonempty class. If B is the event that every label occurs, the union bound gives $\Pr(B^c) \leq M(1 - 1/M)^d \leq Me^{-d/M} =: q_{\mathrm{empty}}$ (at $M = 1$ the first value is zero). Thus the last theorem and B hold simultaneously with probability at least $1 - d^{-2} - q_{\mathrm{empty}}$. When $q_{\mathrm{empty}} < 1$, its conditional failure probability is at most $d^{-2}/(1 - q_{\mathrm{empty}})$; otherwise use the exact positive $\Pr(B)$ as denominator. In the growing range $M\eta_d \leq 1$, the

empty-label bound tends to zero, since $M \le \sqrt{d/(6\log d)}$ and hence $Me^{-d/M} \le \sqrt{d}\,e^{-\sqrt{6d\log d}}$ eventually. Every unlabeled nonempty partition has exactly $M!$ labelings, so the same conditional fraction holds for uniform unlabeled partitions. The formal quotient uses equality of label fibers; it constructs the unique relabeling permutation and proves the uniform-fiber identity. The conditioning statements and the growing-range limit are formalized in `CapacityMaskUnlabeled.lean` and `CapacityMaskGrowth.lean`. Together with the universal difference-cover guarantee, this makes the order $M^{-1/2}$ sharp for that direct certificate method in the stated growing range. It is not an optimal constant, an attained code rate, or a restriction on the stronger nested methods. Final flooring changes a normalized direct certificate by less than $1/N$.

As another comparison, putting $\eta = 1/6$ in the binary change-count event makes every direct two-shift certificate at least $2N/3$, outside probability $(d-1)e^{-d/72}$. Every direct set of at least three shifts already has that lower bound from its ambient term. For any fixed $0 < \varepsilon < 2/3 - 41473/65536$ and sufficiently large d, the run-spectrum bound is strictly below this entire direct envelope, except on probability at most $e^{-128\varepsilon^2 d/529} + (d-1)e^{-d/72}$. For rounded bounds choose d large enough that the difference exceeds one dimension. No comparison here extracts a shared code from a native computation.

D Extension-linear residuals with binary-linear decoders

This section concerns scalar binary entries, so $b = 1$. Put $E = \mathbb{F}_{2^d}$ and $\Phi(x) = x^2$. Use the reversed normal basis from the interval construction to identify a binary row with E; literal column rotation then acts on every row by Φ. No prime or coprimality hypothesis on $d \ge 1$ is needed. Linearity of a decoder below means $\mathbb{F}_2$-linearity on the complete binary view, not E-linearity.

Fix a partition $p : \mathbb{Z}/d \to \mathcal{J}$ into M nonempty classes, and a representative r_J in every class. For a binary linear source C, let $C_0 = C \cap \ker P$, where P is the actual class projection from the completion theorem. Thus every representative row vanishes on C_0. The additional hypothesis is that the image $K \subseteq E^d$ of *this intersection*, in the fixed normal basis, is an E-subspace. It is not a hypothesis on the image $(I - P)C$, nor on C or the baseline separately. Define

$$S = \{c : \exists x \in K,\ x_c \neq 0\}, \qquad a = |S|, \quad a_J = |S \cap p^{-1}(J)|, \quad k = \dim_E K.$$

Lemma D.1 (Frobenius components)**.** Lean: `CapacityFieldLinear.phaseBasis`, `CapacityFieldLinear.sum_components`, `CapacityFieldLinear.components_unique`, `CapacityFieldLinear.component_of_scalarTest`, `CapacityFieldLinear.component_zero_of_scalarTest_zero` [`CapacityFieldLinearComponents.lean`]

For every binary-linear map $f : E^S \to E$ there are unique E-linear functionals $\ell_t : E^S \to E$, indexed by $t \in \mathbb{Z}/d$, such that

$$f(y) = \sum_t \Phi^t(\ell_t(y)).$$

If $K \le E^S$, $c \in S$, and $f(\Phi^c x) = x_c$ for all $x \in K$, then on $V_c = \Phi^c K$ the functional ℓ_{-c} is the coordinate $y \mapsto y_c$ and all other components vanish. For any phase $c \in \mathbb{Z}/d$, if $f(\Phi^c x) = 0$ for every $x \in K$, all components vanish on that view.

Outline of the formal proof. The d distinct automorphisms Φ^t are linearly independent over E. The space $\mathrm{Hom}_{\mathbb{F}_2}(E, E)$ has E-dimension d, so they form a basis. Expand the binary-linear map $\lambda \mapsto f(\lambda y)$ in this basis. Its coefficient at t, followed by Φ^{-t}, is $\ell_t(y)$. Uniqueness of coordinates

proves additivity and $\ell_t(\lambda y) = \lambda \ell_t(y)$, hence E-linearity. Evaluating at $\lambda = 1$ gives the displayed decomposition and its uniqueness.

For $y = \Phi^c x \in V_c$, every λy is still in V_c because K is E-linear. The decoder identity therefore gives

$$\sum_t \Phi^t(\lambda)\Phi^t(\ell_t(y)) = \Phi^{-c}(\lambda)\Phi^{-c}(y_c).$$

Independence of the automorphisms as functions of λ forces the claimed coefficient identities. This argument uses the actual row demand; it supplies no receiver index to the decoder. □

Theorem D.2 (Active-row bound)**.** Lean: `CapacityFieldLinear.active_class_residual_bound`, `CapacityFieldLinear.original_source_class_bound`, `CapacityFieldLinear.finrank_encodedResidual` [`CapacityFieldLinearResidual.lean`]

Suppose C has total binary-linear shared decoders and its actual residual K is E-linear as above. Then, for every class J,

$$a_J + k \le a, \qquad D + d a_J \le Md + da, \qquad D = \dim_{\mathbb{F}_2} C. \tag{23}$$

Consequently

$$D \le Md + d\big(a - \max_J a_J\big) \le Md + d(1 - 1/M)a.$$

Outline of the formal proof. Restrict inputs to their active rows S, extending by zero elsewhere. This gives an E-linear isomorphism between K and an E-subspace of E^S. Restrict each actual decoder by this zero extension and the normal basis identification. For $c \in S \cap p^{-1}(J)$ take its component $L_c = \ell_{J,-c}$. By Lemma D.1, L_c equals the cth coordinate on V_c and vanishes on every V_e with $e \in p^{-1}(J)$, $e \neq c$. The coordinate is nonzero on V_c because c is active. A relation $\sum_c u_c L_c = 0$, restricted successively to these views, therefore has every $u_c = 0$.

The representative row is zero, so all L_c annihilate V_{r_J}. The latter has E-dimension k in an ambient of dimension a; its annihilator has dimension $a - k$. The independent family has size a_J, giving the first inequality. In Lean the annihilating functionals are conjugated by the representative Frobenius automorphism, so they all lie in the annihilator of K itself.

The normal-basis equivalence and the field tower give $\dim_{\mathbb{F}_2} C_0 = dk$. Rank–nullity for $P|_C$, whose image has dimension at most Md, gives $D \le Md + dk$. This proves the second inequality without assuming that the baseline is contained in C. Take the largest a_J for the first consequence; since $\sum_J a_J = a$, its largest term is at least a/M. □

Corollary D.3 (One and two classes)**.** Lean: `CapacityFieldLinear.one_class_residual_eq_bot`, `CapacityFieldLinear.one_class_original_bound`, `CapacityFieldLinear.two_class_original_bound`, `CapacityFieldLinear.two_class_half_bound` [`CapacityFieldLinearCorollaries.lean`]

Under the same hypotheses, one class gives $K = 0$ and $D \le d$. For two nonempty classes J_0, J_1,

$$D \le d\big(1 + \min(|J_0|, |J_1|)\big), \qquad 2D \le d^2 + 2d.$$

Outline of the formal proof. For one class $a_J = a$, so $k = 0$. For two classes $a = a_{J_0} + a_{J_1}$, and the active-row bound gives $k \le \min(a_{J_0}, a_{J_1})$. Each class contains its zero representative, so $a_J \le |J| - 1$. Substitute this into $D \le 2d + dk$ and then use $|J_0| + |J_1| = d$. □

This excludes above-half asymptotic rates in this construction class with binary-linear decoders. Deterministic decoders for extension-linear graph codes can be nonlinear, so those constructions do not contradict the one-class conclusion. The theorem does not cover arbitrary binary sources, alternative row identifications, or $b > 1$, and it supplies no native extraction or implementation claim.

E Measured structural restrictions

The parameters in this appendix describe an actual source subspace. They are not running-time invoices. Neither result supplies its own small-parameter hypothesis for an arbitrary tape computation.

E.1 A robust bound for almost field-linear residuals

Fix the reversed normal-basis identification of each binary row with $E = \mathbb{F}_{2^d}$. Thus column rotation is coordinatewise Frobenius Φ on the d row vectors in E^b. For a two-class code C, let P be the representative-row projection of Section 4, and put

$$L = C \cap \ker P, \quad H = \operatorname{span}_E L, \quad \rho = \dim_{\mathbb{F}_2} H - \dim_{\mathbb{F}_2} L.$$

The common E scalar acts on all rows and lanes; no independent-lane decomposition is assumed. Let a be the number of potentially nonzero residual rows, so $a \le d-2$.

Theorem E.1 (Scalar-closure defect)**.** `Lean`: `CapacityClosure.residual_finite_bound`, `CapacityClosure.capacity_bound`, `CapacityClosure.rounds_eq_floor`, `CapacityClosure.error_ratio_tendsto`, `CapacityClosure.uniform_eventual_bound` [`CapacityClosureAsymptotic.lean`]

For $d \ge 2$, $b \ge 1$, arbitrary deterministic shared decoders, and every integer $R \ge 1$, set

$$\mathcal{E}_R(d) = \frac{d}{10R} + \frac{7\sqrt{d}}{5R}\frac{2^{R/2}-1}{\sqrt{2}-1}.$$

Then

$$\dim_{\mathbb{F}_2} L \le bd\left(\frac{a}{2} + \frac{d}{10} + \mathcal{E}_R(d)\right) + \frac{13}{5}\rho,$$
$$\dim_{\mathbb{F}_2} C \le \frac{3}{5}N + bd + bd\mathcal{E}_R(d) + \frac{13}{5}\rho.$$

Taking $R = \lfloor \frac{1}{2}\log_2 d \rfloor$ for $d \ge 4$ gives $\mathcal{E}_R(d) = o(d)$. In particular $\rho = o(N)$ implies rate at most $3/5 + o(1)$, uniformly over all widths and all lane coupling.

Outline of the formal proof. For a shift set F containing zero write $H_F = \bigcap_{e\in F} \Phi^{-e}H$ and $L_F = \bigcap_{e\in F}\Phi^{-e}L$. The first is E-linear. The binary map from H_F to the direct sum of the quotients $\Phi^{-e}H/\Phi^{-e}L$ has kernel L_F, so

$$\operatorname{codim}_{\mathbb{F}_2}(L_F, H_F) \le |F|\rho. \tag{24}$$

Delete loops from the same-class link graph for F. One joint binary-linear covariance map is

$$B_F(X) = (X_v - \Phi^{v-u}X_u)_{u\to v},$$

with one orientation for each edge. Actual decoder equality makes this map zero on L_F. Hence its rank on H_F is at most $|F|\rho$.

Choose a normal element $\lambda \in E$, so $\lambda \ne \Phi^\delta\lambda$ for every nonzero $\delta \in \mathbb{Z}/d$. Multiplication by λ permutes H_F, so $B_F(\lambda X)$ has the same rank on that space. For an edge of displacement δ, the two observations $A = X_v - \Phi^\delta X_u$ and $A' = \lambda X_v - \Phi^\delta(\lambda)\Phi^\delta X_u$ give

$$A' - \lambda A = (\lambda - \Phi^\delta\lambda)\Phi^\delta X_u.$$

They recover both entire endpoint vectors. Therefore the projection onto all nonisolated rows has binary rank t_F satisfying

$$t_F \le 2|F|\rho. \tag{25}$$

This uses one pair of joint maps, not a sum of edge ranks. It does not assert decoder correctness on $H_F \setminus L_F$.

Let $n_F = bd|S \cap \mathrm{Iso}(F)|$, where S is the active row set. Projection gives $\dim_{\mathbb{F}_2} H_F \le n_F + t_F$ and, for any shift s,

$$\dim_{\mathbb{F}_2}(H_F + \Phi^{-s} H_F) \le n_F + 2t_F.$$

The two error images have separately paid dimensions t_F; neither need be invariant. For a nested chain $F_i = F_{i-1} \cup (F_{i-1} + s_i)$, $F_0 = \{0\}$, Grassmann and this inequality yield

$$\dim_{\mathbb{F}_2} L \le \sum_{i<j} \frac{n_{F_i}}{2^{i+1}} + \frac{n_{F_j}}{2^j} + (2j-1)\rho.$$

Indeed unrolling contributes $\sum_{i=1}^{j} t_{F_i}/2^i \le 2j\rho$ by (25), and $\dim_{\mathbb{F}_2} L = \dim_{\mathbb{F}_2} H - \rho$ subtracts the original defect once.

For $F_1 = \{0, s\}$ and $F_2 = \{0, s, 3s, 4s\}$ the full isolation counts obey

$$I_1 \le Q_s - \tfrac{1}{2} Q_{2s} + g_s, \qquad I_4 \le R_4(s) + 2g_s.$$

For the first assertion, on a nonconstant s-cycle the isolated vertices are precisely singleton runs, counted by $Q_s - R_2(s)$; constant or one-point cycles are covered by g_s. For the second, a vertex isolated in the radius-four graph must be a singleton run whose following opposite-color run has length at least four; these following runs are distinct. Cycles of length at most eight have at most two isolated vertices, paid by $2g_s$. Use Lemma 3.3 and its weaker budget $Q_s + R_2 + 2R_4 \le d$. The one- and two-stage inequalities become

$$\begin{aligned} \dim L &\le bd(a/2 + Q_s/2 - Q_{2s}/4 + g_s/2) + \rho, \\ \dim L &\le bd(a/2 + d/8 + Q_s/8 - 3Q_{2s}/16 + 3g_s/4) + 3\rho. \end{aligned}$$

Their convex combination with weights $1/5, 4/5$ is

$$\dim L \le bd\,(a/2 + d/10 + (Q_s - Q_{2s})/5 + 7g_s/10) + 13\rho/5.$$

Average $s = 2^r u$ for uniform $u \in \mathbb{Z}/d$ and $0 \le r < R$. The Q terms telescope; the initial average is at most $d/2$ and the terminal average is nonnegative. The gcd estimate proved in Section 3, $\mathbb{E}\, g_{2^r u} \le 2\sqrt{d}\, 2^{r/2}$, gives exactly $\mathcal{E}_R(d)$. Finally $\dim C \le \dim L + 2bd$ and $a \le d-2$ prove the second bound. For the stated R, both terms of $\mathcal{E}_R(d)/d$ tend to zero. □

At $\rho = 0$ the theorem covers genuinely E-linear residuals with arbitrary deterministic decoders. It improves the general $41/64$ asymptotic bound whenever $\rho/N < 1/64$ by a fixed margin. The parameter can nevertheless be extensive for the cheapest partial transposer.

Proposition E.2 (Symmetric-copy control)**.** Lean: `CapacityClosure.symmetric_isSharedCode`, `CapacityClosure.scalarHull_supportedSymmetric`, `CapacityClosure.symmetric_copy_defect` [`CapacityClosureControl.lean`]

On the lane-wise symmetric source, the identity output is correct for every receiver partition. For two distinct representative rows its residual satisfies

$$\dim L = \frac{b(d-2)(d-1)}{2}, \qquad \dim_{\mathbb{F}_2} H = bd(d-2), \qquad \rho = \frac{b(d-2)(d+1)}{2}.$$

Thus $\rho \sim N/2$, although straight streaming copy realizes this partial correctness set.

Outline of the formal proof. Zeroing representative rows of a symmetric matrix also zeroes their columns. Restriction to the remaining rows identifies the residual with symmetric matrices of side $d-2$. Each remaining diagonal unit has a nonzero normal-basis coordinate. Extension-field multiples of these units span every row and lane of the zero-representative ambient. The dimension difference gives the displayed defect. The native realization is the literal streaming copy operation; the formal statements establish its symmetric correctness set and the exact algebraic dimensions. □

E.2 Retaining symmetric directions

Let $\mathcal{S} = \operatorname{Fix}\tau$, put $\mathcal{S}_0 = C \cap \mathcal{S}$, and write

$$s_0 = \dim \mathcal{S} = bd(d+1)/2, \qquad \eta = s_0 - \dim \mathcal{S}_0.$$

The following restriction uses actual demanded rows and no field-extension structure. In particular it holds over every field and for nonlinear decoders.

Theorem E.3 (Symmetric-retention bound)**.** Lean: `IsSharedCode.retention_bound` [`CapacityRetentionFinite.lean`]

Choose $k \geq 0$ shifts t_i, let $g_i = \gcd(d, t_i)$, and let $U = \{c : p(c) \neq p(c+t_i)$ for every $i\}$, $u = |U|$. Then

$$D \leq s_0 + (2k-1)\eta + \frac{bd}{2}\sum_i (g_i - 1) + \frac{b}{2}u(u-1). \tag{26}$$

Outline of the formal proof. Put $A(X) = X - X^{\mathsf{T}}$ and $Z_t = \{c : p(c) = p(c+t)\}$. If $X = T^t Y \in C$ with $Y \in \mathcal{S}_0$, the same-view equations give $\operatorname{row}_c X = \operatorname{row}_{c+t} Y$ for $c \in Z_t$. Symmetry of Y then gives $X_{c,j} = X_{j,c}$ in every lane. The restriction of A to rows Z_t therefore vanishes on both $\mathcal{S}_0$ and $C \cap T^t\mathcal{S}_0$. Grassmann and rank–nullity bound its rank by $N - \dim(\mathcal{S}_0 + T^t\mathcal{S}_0)$.

Stacking the k row maps costs at most the sum of their ranks; no independence is assumed. An antisymmetric image with all covered rows zero is supported on $U \times U$, has zero diagonal, and is determined by $bu(u-1)/2$ coordinates in every characteristic. Since $\ker(A|_C) = \mathcal{S}_0$, this proves the stronger exact inequality

$$D \leq \dim \mathcal{S}_0 + \sum_i [N - \dim(\mathcal{S}_0 + T^{t_i}\mathcal{S}_0)] + \frac{b}{2}u(u-1). \tag{27}$$

It is formalized as `IsSharedCode.intrinsic_retention_bound`.

The remaining estimate uses the actual symmetric spaces. For $g = \gcd(d, t)$, the two coordinate symmetries generate the dihedral action whose rotations send (i, j) to $(i + at, j - at)$ and whose reflections send it to $(j - at, i + at)$. Exactly g rotations fix all d^2 coordinates; the other rotations

fix none. Each of the d reflections fixes d coordinates. The space of invariant coordinate functions is the function space on the orbits, so Burnside's formula gives

$$\dim(\mathcal{S} \cap T^t\mathcal{S}) = bd(g+1)/2.$$

This counting proof does not divide by the group order in the field and remains valid in characteristic two. Grassmann, and replacing each symmetric summand by its codimension-η subspace, give

$$N - \dim(\mathcal{S}_0 + T^t\mathcal{S}_0) \le 2\eta + bd(g-1)/2.$$

Substitution in (27) proves the theorem. □

Corollary E.4 (Averaged retention)**.** Lean: `IsSharedCode.retention_average`, `IsSharedCode.retention_normalized` [`CapacityRetentionAverage.lean`]
Let the class sizes be n_J, put $a_d = d^{-1}\sum_t \gcd(d,t)$, and let $k \ge 1$. Then

$$D \le s_0 + (2k-1)\eta + \frac{bdk}{2}(a_d - 1) + \frac{b}{2}\left[\sum_J n_J(1 - n_J/d)^{k/2}\right]^2. \tag{28}$$

Consequently, with $h = \eta/N$ and $\delta = D/N - (d+1)/(2d)$,

$$\delta \le (2k-1)h + \frac{k(a_d-1)}{2d} + \frac{2M^2}{(k+2)^2}. \tag{29}$$

Outline of the formal proof. Average over the full set of k-tuples of shifts. A row in a class of size n_J is uncovered for exactly $(d-n_J)^k$ tuples. The finite Cauchy–Schwarz inequality bounds the sum of squared uncovered-row counts by the square of the sum of square roots of these marginal counts. Grouping rows by their classes and using $u(u-1) \le u^2$ gives the last term of (28); no independence between row events is assumed. Each shift coordinate has the uniform marginal, so its mean gcd cost is exactly $a_d - 1$. For $0 \le x \le 1$, Bernoulli's inequality gives $x(1-x)^{k/2} \le 2/(k+2)$. Applying it to every class and dividing by N proves (29). □

Corollary E.5 (A fixed rate gain requires a fixed loss)**.** Lean: `RetentionAverage.stability`, `RetentionAverage.stability_bounded_classes`, `RetentionAverage.fixed_gain_requires_loss` [`CapacityRetentionStability.lean`]
For fixed M, $\eta = o(N)$ implies $D \le (1/2 + o(1))N$ as $d \to \infty$, uniformly in the field and width. More generally this holds when $M\eta/N \to 0$ and $M(a_d-1)/d \to 0$. For fixed M and $0 < \varepsilon \le 1/2$, if $\liminf D/N \ge 1/2 + \varepsilon$, then

$$\liminf \eta/N \ge \varepsilon^{3/2}/(12M).$$

Outline of the formal proof. The divisor bound $a_d \le 2\sqrt{d}$ makes the gcd term vanish for fixed k. The formalization permits the field, lane space, dimension and class set to vary along the sequence, retaining an actual shared code at every index. For growing M, take $k = \lceil Mt \rceil$ with a fixed real $t \ge 1$. The normalized upper bound is at most

$$\frac{1}{2d} + 4tM\left[\frac{\eta}{N} + \frac{a_d - 1}{2d}\right] + \frac{2}{t^2}$$

above one half. First let $d \to \infty$ and then let $t \to \infty$. For the quantitative statement take $k = \lceil 2M/\sqrt{\varepsilon} \rceil$. The last term is at most $\varepsilon/2$, and $k \le 3M/\sqrt{\varepsilon}$. Take a subsequence attaining the lower limit of η/N and pass the finite bound to the limit; the rate lower limit still applies on that subsequence. This gives the stated constant. Both rates lie in $[0,1]$, so all lower limits used here have explicit boundedness hypotheses in Lean. □

Corollary E.6 (Coprime partners and exact retention)**.** Lean: `IsSharedCode.full_symmetric_retention`, `IsSharedCode.eq_symmetric_of_at_most_one_exception`, `IsSharedCode.prime_full_symmetric_retention` [`CapacityRetentionCoprime.lean`]

Suppose $C \supseteq \mathcal{S}$. Let V be the rows c with no same-class partner $c+t$ for which $\gcd(d,t)=1$, and put $v=|V|$. Then, in every positive dimension,

$$D \le s_0 + b\binom{v}{2}.$$

In particular $v \le 1$ implies $C=\mathcal{S}$. If d is prime, V is exactly the set of rows in singleton classes. Thus a partition into two nonsingleton classes forces $C=\mathcal{S}$.

Outline of the formal proof. Use all coprime shifts in Theorem E.3. Then $\eta=0$, every gcd cost is zero, and the uncovered rows are exactly V. For $v \le 1$ the dimension bound and $\mathcal{S} \subseteq C$ give equality. In prime dimension the nonzero shifts are precisely the coprime shifts. □

Proposition E.7 (Paired-lane control)**.** Lean: `RetentionControl.laneSwap_eq_transpose_iff`, `RetentionControl.paired_source_rate_and_loss` [`CapacityRetentionControl.lean`]

For any field and even entry width b, swapping each paired group of lanes entrywise is transpose-correct exactly on the source consisting of lane pairs (A, A^{T}). This source has

$$D = N/2, \qquad \eta = bd(d+1)/4.$$

Outline of the formal proof. The source is the graph of transposition on one member of each lane pair, so it is linearly equivalent to an arbitrary matrix with $b/2$ lanes. Its intersection with the symmetric space consists of pairs (A, A) with A symmetric. The two explicit linear equivalences give the dimensions. The formalization allows vector-valued lane groups. □

Over the binary alphabet, one fixed finite-state stream transducer realizes the adjacent-pair interchange by buffering two input bits and emitting them in reverse order. It repeats without computing the entry width. For every even width its correctness source is Proposition E.7, with $\eta/N = (d+1)/(4d) \to 1/4$; infinitely many canonical widths are even. This written implementation refutes an unqualified small-loss time supplier. It does not address a supplier restricted to fixed excess above one half.

F Why a large linearly decoded subcode is not free

The source remains a binary linear subspace throughout this appendix. Only decoder linearity changes. The bounds allow a candidate subcode and its decoder to mix all lanes; lane-wise upper bounds are not summed without a joint argument.

Lemma F.1 (Projection with coupled lanes)**.** Lean: `IsLinearSharedCode.inr_comap`, `linear_subcode_lane_bound` [`CapacityLinearization.lean`]

Let G, H occupy disjoint lane groups with the same receiver partition. Suppose every linearly decoded subcode of H has dimension at most L_H. Every linearly decoded subcode $C' \le G \oplus H$ has dimension at most $\dim G + L_H$, even with arbitrary cross-lane linear decoding.

Outline of the formal proof. Project C' onto the G coordinates. Its image has dimension at most $\dim G$. Its kernel is a subcode of H; restrict each full linear decoder to zero G input and project its output to the H coordinates. This is a legal linear shared decoder on the kernel. Rank-nullity gives the assertion. No decoder on the projected G image is needed. □

Lemma F.2 (A width-stable triple obstruction)**.** Lean: `CapacityTriple.exists_width_stable_hard_source` [`CapacityTripleBinary.lean`]

Let $d = 3s$ be odd. Partition $\{0, \dots, s-1\}$ and enlarge each class by its translates by s and $2s$. There is a positive integer ℓ such that, for every width B divisible by ℓ, a binary linear source H_B of dimension $Bd^2/3$ has deterministic shared decoding, while every binary-linearly decoded subcode has dimension at most $2Bd^2/9$.

Outline of the formal proof. Choose $E = \mathbb{F}_{2^\ell}$ containing a primitive d-th root ζ; such an extension exists since d is odd. Write $e_k(t) = \zeta^{kt}$, and let the allowed row space be

$$R = \operatorname{span}_E\{e_{3j+1} + e_{3j+2} : 0 \le j < s\}.$$

All d rows vary independently in R. The resulting binary source has dimension $\ell ds = \ell d^2/3$. For distinct phases c, e, equality between a vector in $\sigma^c R$ and one in $\sigma^e R$ compares the two coefficients in each disjoint frequency pair. Their ratio is respectively ζ^c and ζ^e. Thus every pair coefficient is zero, and the two row spaces intersect only in zero. The entire nonzero matrix views are therefore disjoint, and the total phase-identifying decoder from the proof of Theorem 4.3 returns the actual demanded row.

Every row in R satisfies $(I + \sigma^s + \sigma^{2s})x = 0$, because the two frequencies in each pair are nonzero modulo three. Thus every source matrix satisfies $(I + T^s + T^{2s})X = 0$. If a subcode has binary-linear shared decoders, applying one decoder at a view $T^a X$, using the three same-class receivers, forces

$$\operatorname{row}_a X + \operatorname{row}_{a+s} X + \operatorname{row}_{a+2s} X = 0 \qquad (0 \le a < s).$$

The joint row-sum map on the whole hard source is surjective onto s independent copies of R: set just the first row of each triple to any chosen vector. Its binary rank is ℓs^2, giving the claimed $2\ell d^2/9$ upper bound. Repeat the field-coordinate blocks to width B. The same joint map is surjective onto all repeated blocks, so its rank scales by B/ℓ even when the subcode and decoders mix those blocks arbitrarily. □

Theorem F.3 (Above-half sources can resist subcode promotion)**.** Lean: `CapacitySubcode.exists_above_half_barrier` [`CapacitySubcodeMixture.lean`]

Lean: `CapacitySubcode.above_half_seed_barrier` [`CapacitySubcodeBarrier.lean`]

For every odd $d \ge 9$ divisible by three, at suitable binary widths and a two-class partition there is a deterministically decoded linear source of rate

$$\frac{1}{2} + \frac{1}{8d+18}$$

whose every linearly decoded subcode has rate at most $\frac{1}{2} - \frac{1}{8d+18}$. More generally, a two-class seed at odd s coprime to three with rate $1/2 + \varepsilon$, $\varepsilon > 0$, gives at $d = 3s$, after a width enlargement, a source of rate $1/2 + \delta$ whose every linearly decoded subcode has rate at most $1/2 - \delta$, where

$$\delta = \frac{\varepsilon}{18\varepsilon + 12}.$$

Outline of the formal proof. For the second statement apply Theorem 5.1 with $m = 3$. The good source has rate $\alpha = 1/2 + e$, $e = \varepsilon/3$, on the same tripled partition used in Lemma F.2. Repeat binary lane blocks to put it and the hard source at a common width B; this repeats codes, and does not extend a nonlinear decoder's field. Choose positive integers g, h with $h/(g+h) = e/(e + 2/9)$, possible since all seed rates are rational. Take g good blocks and h hard blocks. Their concatenated

deterministic decoders work on the direct sum. Its rate is $(1-t)\alpha + t/3$; by the joint-width hard bound and Lemma F.1, every linearly decoded subcode has rate at most $(1-t)\alpha + 2t/9$. Substitution of $t = e/(e+2/9)$ gives the stated symmetric offsets $\delta = e/[18(e+2/9)]$.

For the unconditional statement use the symmetric good source, of rate $1/2 + 1/(2d)$, at the tripled two-class partition. Take $g = 4d$ good blocks and $h = 9$ hard blocks. The two rates are respectively $(2d+5)/(4d+9)$ and at most $(2d+4)/(4d+9)$, which are the displayed offsets from one half. □

The unconditional excess tends to zero. A family with a fixed positive excess in the second statement is conditional on such a seed family; it is not supplied here. Nor does the theorem separate the maximum capacities over unrelated codes: its conclusion concerns subcodes of one actual source. The enlarged widths can exceed both one and the canonical logarithmic width, so neither restricted-width conclusion is inferred.

G A reflection barrier at every binary word width

The triple-relation family gives a direct linearization obstruction at its specified cyclic dimensions and extension-compatible widths. The following family has a different scope: every binary width $b \geq 1$, odd prime side length, and source rate tending to one half. It does not replace either the above-half subcode construction or the triple relation-rank diagnostic.

Let $d = 2m+1$ be an odd prime, and put

$$W_b = \{x \in (\mathbb{F}_2^b)^d : x_0 = 0,\ x_j = x_{-j}\ \forall j\}, \qquad C_b = \{X : \text{every row of } X \text{ belongs to } W_b\}.$$

All rotations act on entry columns and leave lane indices fixed.

Theorem G.1 (The reflection source)**.** Lean: `CapacityReflection.finrank_rowSpace`, `CapacityReflection.finrank_source`, `CapacityReflection.source_disjoint_views`, `CapacityReflection.source_shared` [`CapacityReflectionSource.lean`]

The source C_b has binary dimension bdm. Its distinct rotated views $T^c C_b$, $T^e C_b$ intersect only in zero. It therefore admits one total deterministic shared decoder for every receiver, and in particular the same function can serve the two nonempty classes $\{0\}$ and $\mathbb{Z}/d \setminus \{0\}$.

Outline of the formal proof. The m pairs $\{j, -j\}$ of nonzero coordinates are free b-vectors, so $\dim W_b = bm$ and $\dim C_b = dbm$. The formal equivalence chooses the smaller positive representative of each pair and treats coordinate zero separately.

A vector in $\sigma^c W_b$ is invariant under reflection $j \mapsto -j - 2c$ and vanishes at $-c$. Two such symmetries with distinct centers give invariance under translation by $2(e-c) \neq 0$. Because d is prime, this translation generates the whole cycle. Thus a vector in two distinct translates is constant, and its zero coordinate forces it to be zero. Apply this independently to every row and all its lanes. The formal proof obtains any displacement as a multiple of the nonzero step in the field $\mathbb{Z}/d$.

For a nonzero received view, disjointness makes its receiver phase and underlying source matrix unique. Return that source's requested row. Return zero at the zero view and choose arbitrary values off the union of the views. This is one well-defined total function. It is not asserted linear or efficiently computable. □

Theorem G.2 (Every-width linear subcode barrier)**.** Lean:
`CapacityReflection.one_class_finrank_le`, `CapacityReflection.two_class_finrank_le`, `CapacityReflection.two_class_codimension_lower`, `CapacityReflection.two_class_positive_deletion`, `CapacityReflection.deletion_rate_limit` [`CapacityReflectionBarrier.lean`]

Let $C' \le C_b$ be any binary-linear subcode; it may couple every row and every lane. If it has a single binary-linear shared decoder, then

$$\dim_{\mathbb{F}_2} C' \le bm(m+1).$$

If instead it has two binary-linear shared decoders for the original classes $\{0\}$ and $\mathbb{Z}/d \setminus \{0\}$, then

$$\dim C' \le b(m(m+1)+d), \qquad \operatorname{codim}_{C_b} C' \ge b(m^2-d). \tag{30}$$

The deletion bound is positive for odd primes $d \ge 7$ and is asymptotic to $bd^2/4$. More generally, these linear-subcode bounds hold over every field of characteristic two and at every odd side length $d = 2m+1$. Primeness is needed for the preceding deterministic-decoder construction.

Outline of the formal proof. First suppose there is one linear decoder L. Extend the coefficient field to a splitting field E for a primitive dth root of unity. The formal construction takes the E-span of the embedded original basis of C' and proves that basis remains independent. The extended code therefore has the same dimension. Extending the actual coordinate coefficients of L preserves its shared-decoder equations, and span induction preserves the literal reflection and zero-coordinate equations defining C_b. All row and lane coordinates are included. This step extends only linear equations; it makes no scalar-extension claim about the deterministic decoder of Theorem G.1.

Since d is odd, its image in the characteristic-two field is nonzero; the formal primitive-character construction supplies the required extension. Choose the Fourier basis $e_a(j) = \zeta^{aj}$, $a \in \mathbb{Z}/d$. Its Vandermonde matrix is invertible. Apply this *same* change of basis to both physical indices, preserving the transpose swap, and leave the lanes unchanged. Let $\mathcal{F}$ be the one-index synthesis and define the column mode $M_j(v)_{r,c,\ell} = \zeta^{jc}(\mathcal{F}v)_{r,\ell}$. Writing X as the two-index synthesis of Z, its actual rotated view is

$$T^c X = \sum_j \zeta^{jc} M_j(\operatorname{col}_j Z).$$

Apply L and compare the independent characters in the receiver c with the demanded row of X. This gives $\mathcal{F}(\operatorname{row}_j Z) = L(M_j(\operatorname{col}_j Z))$. Consequently, with the fixed linear maps $B_j = \mathcal{F}^{-1} L M_j$, agreement takes the form

$$\operatorname{row}_j(Z)^{\mathsf{T}} = B_j \operatorname{col}_j(Z), \qquad j \in \mathbb{Z}/d, \tag{31}$$

where $B_j : E^{db} \to E^{db}$ may mix every row and lane. No frequency equation or dimension certificate is supplied as a premise: the formal proof derives these equations from the original decoder.

In the Fourier basis a reflected row has equal coefficients at $a, -a$: reflection exchanges those characters, and their independence forces coefficient equality. Evaluating the row at physical coordinate zero sums all coefficients. Equal nonzero pairs cancel in characteristic two, so the zero-coordinate condition forces coefficient zero to vanish. Conversely these coefficient conditions imply both original conditions. Consequently

$$\operatorname{col}_0(Z) = 0, \qquad \operatorname{col}_a(Z) = \operatorname{col}_{-a}(Z).$$

The further Fourier change on the row index preserves these column identities. Equation (31) at zero now forces $\operatorname{row}_0(Z) = 0$ as well.

Choose representatives $1, \dots, m$ of the nonzero frequency pairs. Group the remaining rows and columns into those pairs. A 2-by-2 block has duplicate columns and is specified by $v_{a,b} \in E^{2b}$, its two entries in column b. The equations for frequencies a and $-a$ use the same input column, and show that the whole compressed row $(v_{a,b})_b$ is a fixed linear function of the compressed column $(v_{c,a})_c$. Equivalently, for fixed linear maps $L_{a,b,c} : E^{2b} \to E^{2b}$,

$$v_{a,b} = \sum_c L_{a,b,c}(v_{c,a}).$$

All cross-lane coefficients remain in these maps.

Supply the strict upper blocks $v_{a,b}$, $a < b$, and all diagonal blocks. There are $\binom{m}{2} + m$ blocks, hence $bm(m+1)$ field coordinates. Recover rows in descending order. In column a, the entries above the diagonal were supplied, the diagonal was supplied, and those below it belong to already recovered rows. The whole column is known and its fixed linear maps determine row a. Thus the supplied-coordinate projection is injective on the agreement space, proving its dimension bound. The formal proof reuses the stronger descending block-recovery theorem, charging each diagonal kernel by at most $2b$.

Finally suppose C' has decoders L_0, L_1 for the two original classes. In the *original binary coordinates* impose $L_1(X) = \mathrm{row}_0(X)$. This is one linear map to a bd-dimensional row, so its kernel cuts C' by at most bd dimensions. On the resulting subcode C'', the single decoder L_1 works also at receiver zero; it already works at every nonzero receiver. The one-class bound gives $\dim C' \le \dim C'' + bd \le bm(m+1) + bd$. Subtract from $\dim C_b = bdm$ and use $d = 2m+1$ to get the stated deletion bound. At $m \ge 3$, $m^2 - (2m+1) > 0$, which covers the odd primes $d \ge 7$. □

The source rate is $(d-1)/(2d)$, approaching one half from below, and the statement covers every b including $b = 1$ and the canonical width $b(d)$. Prime side length is used for the disjointness of rotated source views; characteristic two is used for the Fourier zero coefficient. The two-class partition in the linear bound is specifically singleton versus complement. The linear bound itself needs only odd side length. The result supplies neither a paid implementation of the nonlinear decoder nor a full-cube or native impossibility theorem.

H A bounded extension to a single-site update

The balance argument also handles the other single-site endpoint of the masked-neighbor family. This extension is included to specify what the pre-release strengthening check establishes and what it leaves open.

Proposition H.1 (One binary neighbor update)**.** Lean: `single_at_most_two_classes_impossible` [`CompositionSingleSiteEndpoint.lean`]

Let $d \ge 3$ and let $L = I + E_{0,1}$ act on every binary row/lane word, where $E_{i,j}x$ has its only possibly nonzero coordinate, x_j, at i. For every positive width, there is no complete factorization $\tau = KBLH$ in which B, H are column-equivariant total maps and K has at most two receiver classes.

Outline of the formal proof. Absorb B into K. We prove adjacent-pair transitivity for $Q = LTL^{-1}$ and then apply Theorem 7.4. Put $U_i = I + E_{i,i+1}$, with cyclic indices. Since $L = U_0 = L^{-1}$, conjugating $\langle Q, TQT^{-1} \rangle$ by L^{-1} gives $\langle T, A \rangle$, where

$$A = U_1 U_0 U_{-1} U_0.$$

For $d \geq 4$, direct multiplication gives $A = I + N$, with

$$N = E_{1,2} + E_{-1,0} + E_{-1,1}, \qquad N^2 = E_{-1,2}, \qquad N^3 = 0.$$

In characteristic two, $A^2 = I + E_{-1,2}$. Conjugating by rotations therefore gives every transvection $I + E_{i,i+3}$.

If $d = 4$, the directed step-three edges form one cycle. The elementary identity

$$[I + E_{i,j}, I + E_{j,k}] = I + E_{i,k} \qquad (i, j, k \text{ distinct})$$

along simple directed paths yields every $I + E_{i,j}$ with $i \neq j$. For $d \geq 5$, use the already obtained $E = I + E_{0,3}$. Here $Ae_0 = e_0 + e_{-1}$, while $e_3^* A^{-1} = e_3^*$, because row three is unchanged by A and its inverse. Thus

$$(AEA^{-1})E = I + E_{-1,3}.$$

Rotations also give every step-four transvection. Within each orbit of the step-three rotation, a simple directed path from i to $i-3$ gives the reverse step-three transvection by repeated commutators. If that orbit has two vertices, the reverse edge is already a generator. Since $d \geq 5$, the indices $i, i+4, i+1$ are distinct, and

$$[I + E_{i,i+4}, I + E_{i+4,i+1}] = I + E_{i,i+1}.$$

The resulting step-one cycle gives every elementary transvection by the same chain argument. Such transvections act transitively on nonzero binary words: preserve one nonzero coordinate and add it to clear every other nonzero coordinate, then rotate.

When $d = 3$, the coincident indices require a separate calculation. The same product is $A(a, b, c) = (a, a + c, a + b + c)$. It sends 100 to 111 and then to 101. Their rotations include the three unit vectors, the three weight-two vectors and the all-one vector, proving transitivity on all seven nonzero words. This completes the adjacent-pair argument for every $d \geq 3$.

Hence the partition of K must properly color the cycle. Odd d is impossible with two colors. For even $d \geq 4$ a proper two-coloring has period two. But $L(1, \ldots, 1) = (0, 1, \ldots, 1)$, so no nonzero column rotation preserves $L\,\mathrm{Fix}(T)$. The period constraint in Theorem 7.6 rules out this last case. □

For clarity, this does not settle the family $L_m = I + \mathrm{diag}(m)T$ with an arbitrary binary mask m. If m has at least one zero, the weighted shift is nilpotent and $L_m^{-1} = \sum_{j=0}^{d-1} (\mathrm{diag}(m)T)^j$. The all-one mask is singular. A period r of m makes L_m commute with T^r and gives the class bound of Theorem 7.2. The masks with exactly one zero and exactly one one are covered by Corollary 7.7 and Proposition H.1, respectively. The remaining aperiodic masks, and half-period masks with two final classes, are not excluded by these arguments. All statements concern complete algebraic maps; no native implementation or physical cost bound is asserted.

I A complete scattered first factor and its failure to compose

A source decoder does not automatically extend to a bijection on all matrices. For the scattered normal-basis source, such an extension does exist. The particular extension below nevertheless forces every receiver of the remaining factor to have its own class. Both assertions concern the complete finite state space; neither supplies an efficient evaluator for the orbit permutation.

I.1 The source and a total permutation

Let $d \geq 2$ and $b \geq 1$, put $V = (\mathbb{F}_2^b)^{d\times d}$, and let $B = (\mathbb{F}_2^b)^d$ be the whole-column alphabet, of cardinality $q = 2^{db}$. Use the reversed normal-basis identification $\mathbb{F}_2^d \cong E = \mathbb{F}_{2^d}$ underlying Theorem 4.3; a column rotation by s acts on a row as Φ^s, where $\Phi(x) = x^2$. Let β be its normal element. Its d conjugates are distinct, and none equals 0 or 1. For $u \in B$, write $a_\ell \in E$ for the field encoding of $(u_{j,\ell})_{j\in\mathbb{Z}/d}$ and define $w(u)$ by its encoded rows:

$$w(u)_{0,\cdot,\ell} = a_\ell, \qquad w(u)_{i,\cdot,\ell} = \beta a_\ell \quad (i \neq 0).$$

Set $W = \{w(u) : u \in B\}$. Also write $\bar{u}$ for the matrix whose every column is u, and let $z(u)$ have column zero equal to zero and all other columns equal to u.

Theorem I.1 (A complete scattered first factor)**.** Lean: `scatteredSource_finrank`, `scattered_orbit_injective`, `scattered_punctured_orbits_disjoint` [`CompositionScatteredSource.lean`]

Lean: `scatteredCompletion_equivariant`, `scatteredFirstFactor_classUniform`, `scatteredFirstFactor_source`, `scatteredFirstFactor_fixed_preimage` [`CompositionScatteredCompletion.lean`]

Lean: `columnPatch_involutive`, `columnPatch_nonzero_column` [`CompositionSparsePatch.lean`]

The binary-linear source W has dimension db. There are total permutations H, R, K_1 of V such that H commutes with column rotation, R changes only column zero, and $K_1 = RH$ is class-uniform for the two classes $\{0\}$ and $(\mathbb{Z}/d) \setminus \{0\}$. They satisfy

$$H(w(u)) = z(u), \qquad K_1(w(u)) = \bar{u} \quad (u \in B), \qquad K_1^{-1}(\mathrm{Fix}(T)) = W.$$

Here R is an involution defined on every input.

Outline of the formal proof. The row-zero projection recovers u, so w is an injective binary-linear map and $\dim W = db$. If $T^s w(u) = T^t w(v)$ with $u, v \neq 0$, choose a nonzero lane parameter. The two row types give

$$\Phi^s(a) = \Phi^t(a') \neq 0, \qquad \Phi^s(\beta)\Phi^s(a) = \Phi^t(\beta)\Phi^t(a').$$

Cancellation and the distinct conjugates of β give $s = t$, then all lane parameters agree. Thus $(s, u) \mapsto T^s w(u)$ is injective for $u \neq 0$. Similarly $(s, u) \mapsto T^s z(u)$ is injective: its unique zero column determines s, and any other column determines u.

These two orbit families are disjoint. Suppose a rotated $w(u)$ equals a rotated $z(v)$. Choose a lane whose row-zero field parameter is nonzero. On the punctured side, a row in this lane is either zero or the common punctured binary word, according as the corresponding bit of v is zero or one. Both rows zero and one on the scattered side are nonzero. They must therefore be equal on the punctured side. In field coordinates this would give $\Phi^s(\beta)\Phi^s(a) = \Phi^s(a)$ with $a \neq 0$, contradicting $\Phi^s(\beta) \neq 1$.

Define H to exchange $T^s w(u)$ and $T^s z(u)$ for every s and every $u \neq 0$, and fix every remaining state. Injectivity and disjointness make these disjoint pairs, so this is an actual total involution. Rotation permutes the pairs with their orientation preserved, hence $HT = TH$. Zero lies outside both families and is fixed. This specializes the completion to the present source and needs no assumed equivariant extension.

Define R to exchange $z(u)$ and $\bar{u}$ for each $u \neq 0$, fixing all other inputs. Each pair agrees outside column zero, so R leaves those columns unchanged. The pairs are disjoint: a punctured state has zero column zero, whereas a nonzero constant state does not. Thus R is again a total involution and $Rz(u) = \bar{u}$, including $u = 0$.

For the exceptional class use the decoder $Y \mapsto \mathrm{col}_0(RH(Y))$; for the other class use $Y \mapsto \mathrm{col}_0 H(Y)$. If $c \neq 0$, equivariance gives $\mathrm{col}_0 H(T^c X) = \mathrm{col}_c H(X) = \mathrm{col}_c K_1(X)$. The exceptional decoder gives column zero by definition. This proves sharedness on every input. Finally the q matrices $\bar{u}$ exhaust $\mathrm{Fix}(T)$, and bijectivity of K_1 gives the stated preimage equality. □

Corollary I.2 (The exceptional class is necessary)**.** Lean: `scattered_one_class_impossible` [`CompositionScatteredCompletion.lean`]

No injective equivariant map sends all of W into $\mathrm{Fix}(T)$.

Outline of the formal proof. Choose $u \neq 0$. Equivariance and the fixed image imply

$$K(Tw(u)) = TK(w(u)) = K(w(u)).$$

Injectivity forces $Tw(u) = w(u)$, contradicting the free-orbit calculation above. A one-class complete map is equivariant, so it is excluded as well. □

I.2 Complete-input disagreement

The agreement estimates in Proposition 7.3 have the following robust form. The comparison need not be a permutation.

Proposition I.3 (Robust comparison with an equivariant map)**.** Lean: `agreement_error_cover`, `robust_full_input_count`, `robust_repeated_class_count` [`CompositionRobustAgreement.lean`]

Let A be any finite alphabet with at least two elements and put $q = |A|^d$. Let $K_1, H \colon A^{d\times d} \to A^{d\times d}$ be total maps, with H equivariant, and let K_2 be class-uniform for a partition p with M nonempty classes of sizes n_J. Under the uniform law on the complete input space, put

$$\gamma = \Pr[K_2K_1(X) \neq \tau X], \qquad \epsilon = \Pr[K_1(X) \neq H(X)].$$

Then

$$\gamma + \epsilon \;\geq\; 1 - \frac{M}{d} - \frac{\sum_J n_J(n_J - 1)}{2dq}.$$

If a receiver class is repeated, also $\gamma + \epsilon \geq (1 - q^{-1})/2$.

Outline of the formal proof. The map K_2H has the same receiver partition, with decoder $f_J \circ H$. If K_2K_1 is correct at X and $K_1(X) = H(X)$, then K_2H is correct there. The entire input space is therefore covered by the correct inputs of K_2H, the failures of K_2K_1, and the disagreements between K_1 and H. Sum their cardinalities and apply the two bounds of Proposition 7.3 to K_2H. The formal statements retain the integer counts and multiply through by the positive denominators. They use no injectivity of H or K_1. □

Theorem I.4 (Sparse completion forces a fully separated second factor)**.** Lean: `columnPatch_support_card` [`CompositionSparsePatch.lean`]

Lean: `scatteredFirstFactor_disagreement`, `scattered_remaining_factor_singletons` [`CompositionSparseObstruction.lean`]

For the binary construction of Theorem I.1,

$$|\{X : R(X) \neq X\}| = 2(q-1), \qquad \Pr[K_1(X) \neq H(X)] = \frac{2(q-1)}{q^d}.$$

If $d \geq 3$ and a class-uniform total map K_2 satisfies $K_2K_1 = \tau$ on all inputs, every receiver class of K_2 is a singleton. Thus its uniquely required remaining factor τK_1^{-1} needs d classes.

Outline of the formal proof. The support of R consists exactly of the two states $z(u), \bar{u}$ for every $u \neq 0$. Bijectivity of H transports this support count to the disagreement count for RH and H. If K_2 had a repeated class, the previous proposition with $\gamma = 0$ would force

$$\frac{2(q-1)}{q^d} \geq \frac{q-1}{2q}, \qquad \text{hence} \qquad q^{d-1} \leq 4.$$

But $q = 2^{db} \geq 8$ when $d \geq 3$ and $b \geq 1$. The formal proof uses the equivalent integer inequalities, avoiding subtraction or division of probabilities. No exclusion at $d = 2$ is asserted. □

This disagreement counts complete input states. It is different from the number of output columns modified by a map, a linear error rank, or a time charge. A change confined to one output column can affect almost every input.

I.3 Endpoint classes are not multiplicative

The need to retain actual factors is already visible in a short bijective control. For binary matrices define rowwise maps

$$(B_1X)_{i,c} = \begin{cases} X_{i,0} + X_{i,1}, & c = 0, \\ X_{i,c}, & c \neq 0, \end{cases} \qquad (B_2X)_{i,c} = X_{i,c} + \sum_j X_{i,j}.$$

Proposition I.5 (Two classes followed by one can require all classes)**.** Lean: `endpointFirst_involutive`, `endpointSecond_involutive`, `endpointFirst_classUniform`, `endpointSecond_equivariant`, `endpoint_composite_getter`, `endpoint_composite_requires_singletons` [`CompositionEndpointBlowup.lean`]

For every even $d \geq 2$, the maps B_1, B_2 are total involutions. The first uses the singleton/rest partition and the second is equivariant. Nevertheless every class-uniform representation of B_2B_1 has d singleton classes.

Outline of the formal proof. The first map adds the unchanged column one twice when squared. For the second, let $s_i = \sum_j X_{i,j}$. The row sum after the map is $s_i + ds_i = s_i$ because d is even in $\mathbb{F}_2$, so its square is also the identity. The sum is rotation-invariant, proving that B_2 is equivariant; the two stated decoders for B_1 are $Y \mapsto Y_{\cdot,0} + Y_{\cdot,1}$ and $Y \mapsto Y_{\cdot,0}$.

The normalized getter of the composite at receiver c is

$$g_c(Y)_i = Y_{i,0} + \sum_j Y_{i,j} + \begin{cases} 0, & c = 0, \\ Y_{i,1-c}, & c \neq 0. \end{cases}$$

These functions are pairwise distinct. Comparing $c = 0$ with $e \neq 0$, use a row supported at $1 - e$. Comparing nonzero distinct c, e, use a row supported at $1 - c$; the extra term at $1 - e$ vanishes. A shared class would make the corresponding normalized getters equal on every input, a contradiction. The formal proof derives this getter equality from the complete class-uniform interface. □

J Balanced fibres beyond transitivity

Transitivity in Theorem 7.4 is a sufficient way to obtain an orbit divisor. It is not necessary. The statements below use an actual invariant scalar-word action and derive the tuple counts through stabilizers. They do not assume transitivity on tuples or independent behaviour of the row and lane actions.

J.1 The fixed-word residue test

Let a group U act on a finite scalar-word set Ω, and let I be a finite index set. The action on Ω^I is diagonal. Write $h = |\operatorname{Fix}_U(\Omega)|$ and $r = |I|$.

Theorem J.1 (Fixed-word residues)**.** Lean: `tuple_orbit_divisor`, `fixedTupleEquiv`, `invariant_fiber_fixed_residue`, `balanced_fixed_word_residue` [`CompositionOrbitResidues.lean`]

Suppose m divides the cardinality of every non-singleton U-orbit in Ω. Let $f\colon \Omega^I \to Z$ be U-invariant, where Z is finite, and suppose every fibre has the same cardinality F. Then

$$|Z|\,(F \bmod m) \le h^r.$$

In particular, in the complete matrix interface over a scalar alphabet of size a, with $r = db$, $|Z| = a^{db}$ and $F = a^{db(d-1)}$, any such invariant action satisfies

$$h^{db} \;\ge\; a^{db}\bigl(a^{db(d-1)} \bmod m\bigr). \tag{32}$$

The integer m need not be prime.

Outline of the formal proof. If a tuple x is not fixed, some coordinate x_i is not fixed. The stabilizer containment $U_x \le U_{x_i}$ shows that $|U \cdot x_i|$ divides $|U \cdot x|$. Therefore every non-singleton tuple orbit has size divisible by m.

An invariant fibre is a union of tuple orbits. Its cardinality modulo m equals the number of its fixed tuples modulo m. Consequently each fibre contains at least $F \bmod m$ fixed tuples. Fixed tuples are precisely tuples of fixed scalar words, of which there are h^r. Summing over the disjoint fibres proves the inequality. In the complete matrix setting, bijectivity gives the stated fibre size, as in the proof of Theorem 7.4. The formal argument proves the class equation for an arbitrary common orbit divisor and constructs the equivalence between fixed tuples and tuples of fixed words. □

Corollary J.2 (Too few fixed words)**.** Lean: `balanced_fixed_word_small_impossible` [`CompositionOrbitResidues.lean`]

Under the preceding hypotheses, suppose $r > 0$, $m > 1$, $\gcd(m, a) = 1$, $|Z| = a^r$, and $F = a^n$ for some $n \ge 0$. If $h < a$, no such invariant balanced function exists. In particular this applies to any prime divisor m coprime to a.

Outline of the formal proof. Coprimality implies $m \nmid a^n$, so $a^n \bmod m \ge 1$. The residue inequality gives $a^r \le h^r$, whereas $h < a$ and $r > 0$ give the strict reverse inequality. □

There is a useful version requiring only one fibre, without counting the fixed set.

Proposition J.3 (A semiregular divisor)**.** Lean: `no_balanced_semiregular`, `no_balanced_free_off_zero` [`CompositionOrbitResidues.lean`]

Suppose U fixes $0 \in \Omega$ and every orbit of a word other than zero has cardinality divisible by $m > 1$, with $\gcd(m, a) = 1$. If $f\colon \Omega^I \to Z$ is invariant, no fibre avoiding the all-zero tuple has cardinality a^n, for any $n \ge 0$. In particular, a cyclic action of order m that is free away from zero has this obstruction. Hence it cannot be an invariance subgroup of a nonconstant balanced decoder whose fibre sizes are powers of a.

Outline of the formal proof. Every tuple in the selected fibre has a nonzero coordinate. Its stabilizer is contained in that coordinate's stabilizer, so its orbit size is a multiple of m. The invariant fibre is a disjoint union of these orbits and also has size divisible by m. This contradicts $\gcd(m, a^n) = 1$. For a free action of a finite cyclic group of order m, each nonzero scalar orbit has size exactly m, giving the stated specialization. The proof uses this orbit-size condition directly; it does not assume that all nonzero scalar words belong to one orbit. □

J.2 An actual nontransitive order-five example

Set $d = 4$ and $\Omega = \mathbb{F}_2^4$, with cyclic shift $(Tx)_i = x_{i+1}$. Define the literal word operation

$$L(x_0, x_1, x_2, x_3) = (x_2,\ x_2 + x_3,\ x_1 + x_2,\ x_0 + x_2), \qquad Q_c = T^c L T L^{-1} T^{-c}.$$

All equalities in the next result concern these displayed operations; no assumed scalar-group identification is used.

Theorem J.4 (Balance excludes a nontransitive adjacent pair)**.** `Lean`: `orderFivePerm_zero`, `orderFivePerm_fixed`, `orderFivePerm_order`, `orderFive_conjugate_difference`, `orderFive_pair_nontransitive`, `orderFive_no_balanced_covariance`, `orderFive_adjacent_class_impossible` [`CompositionOrderFive.lean`]

The operation L is a zero-fixing permutation of order five, and fixes no other word. The group $\langle Q_0, Q_1\rangle$ is not transitive on the fifteen nonzero words. Nevertheless, for every $b \geq 1$, a complete factorization $KLH = \tau$ on $\mathbb{F}_2^{4\times 4\times b}$, with H equivariant and K class-uniform, must satisfy $p(0) \neq p(1)$. The same scalar L is applied independently to every row and lane; H and K may be nonlinear and may mix all lanes.

Outline of the formal proof. Substitution gives

$$L^4 x = (x_0 + x_3,\ x_0 + x_2,\ x_0,\ x_0 + x_1), \qquad L(L^4 x) = L^4(Lx) = x.$$

The fixed-point equations give successively $x_0 = x_2$, $x_1 = 0$, $x_3 = 0$, and $x_2 = 0$. Thus L fixes only zero, and since $L^5 = I$ and $L \neq I$, its order is five. The actual conjugates are

$$Q_0 x = (x_0 + x_1,\ x_1 + x_3,\ x_1,\ x_1 + x_2),$$
$$Q_1 x = (x_0 + x_2,\ x_0,\ x_0 + x_1,\ x_0 + x_3).$$

They satisfy

$$Q_0 Q_1^{-1} = L.$$

Write e_i for the coordinate unit words and $v = (1, 1, 1, 1)$. On $\{e_0, e_1, e_2, e_3, v\}$, the first conjugate fixes e_0 and cycles (e_1, v, e_2, e_3); the second fixes e_3 and cycles (e_0, v, e_1, e_2). Thus both preserve this five-word set. In particular no group element sends a unit word to a word with exactly two nonzero coordinates, so the pair group is not transitive on the nonzero words. Lean checks these finite identities on the sixteen literal words in its kernel.

If $p(0) = p(1)$, the common shared decoder f has $f(Q_0 Y) = \sigma f(Y) = f(Q_1 Y)$. Substitute $Q_1^{-1} Y$ to obtain $f(LY) = f(Y)$, and hence invariance under the cyclic group generated by L. Because L has prime order five and fixes only zero, every nonzero scalar orbit has size five. Stabilizer containment therefore makes every nonzero tuple orbit size divisible by five.

Exactness on the complete finite space makes H and K bijective. The column decoder is balanced: each value has exactly 2^{12b} preimages in the $4b$-tuple domain. Choose an output value different from $f(0)$; it exists because the column alphabet has $2^{4b} > 1$ values. Its fibre avoids zero and is invariant under L, so the semiregular-divisor proposition says its size is divisible by five. The power 2^{12b} is not, a contradiction. The formal endpoint derives covariance and balance from the full product identity; neither property is supplied as a separate certificate. □

These tests are necessary conditions for an actual complete factorization. Passing an orbit-assignment test for one shared function does not produce jointly independent output columns or a bijective whole-matrix map. The arguments also impose no cost bound on evaluating a scalar-word permutation or an orbit representative.

J.3 Field multipliers and two scope checks

Proposition J.5 (Multiplier class costs)**.** Lean: `fieldMultiplierConjugate_eq`, `multiplier_class_residue`, `multiplier_class_count`, `multiplier_class_count_prefix` [`CompositionMultiplierClasses.lean`]
Lean: `actual_multiplier_getters_card`, `multiplier_degree_le`, `multiplier_binary_degree_eq` [`CompositionMultiplierGetters.lean`]

In the balanced-intertwiner interface, identify scalar words over $\mathbb{F}_q$ with $E = \mathbb{F}_{q^d}$ in a reversed normal basis, so rotation is $\Phi(x) = x^q$. Let $L(x) = \lambda x$, $\lambda \neq 0$, on each row and lane. Put $\kappa = \lambda^{1-q}$, $h = [\mathbb{F}_q(\kappa) : \mathbb{F}_q]$ and $h_0 = [\mathbb{F}_q(\lambda) : \mathbb{F}_q]$. Every second-layer class is contained in one residue class modulo h, so it requires at least h classes. More generally, among the first n receivers there are at least $\min(n, h)$ distinct class labels, for every $0 \leq n \leq d$. The complete operation L itself requires exactly h_0 distinct class getters, and $h \leq h_0$. For $q = 2$, $h = h_0$.

Outline of the formal proof. Writing m_z for scalar multiplication, direct composition gives

$$Q_c = m_{\Phi^c(\kappa)}\Phi, \qquad Q_c Q_e^{-1} = m_{\Phi^c(\kappa)/\Phi^e(\kappa)}.$$

If c, e share a decoder, its covariance makes it invariant under the second map. A multiplier $\mu \neq 1$ has order $t > 1$ dividing $q^d - 1$; its cyclic action is free away from zero. Since $\gcd(t, q) = 1$, Proposition J.3 contradicts the required balance. Thus $\Phi^c(\kappa) = \Phi^e(\kappa)$, equivalent to $c \equiv e \pmod{h}$. The first $\min(n, h)$ receivers have distinct residues, proving the prefix bound without requiring a full extension-degree set of receivers.

Let $\ell_0 : E \to \mathbb{F}_q$ extract coordinate zero. The unique full-domain getter for L at receiver c is, on each row and lane, $x \mapsto \ell_0(\Phi^c(\lambda)x)$: rotate the received word back, apply L, then rotate its demanded coordinate to zero. Two such maps agree exactly when their scalar multipliers agree, since a nonzero scalar difference is invertible and $\ell_0 \neq 0$. There are h_0 conjugates, giving both necessity and sufficiency. The inclusion $\kappa \in \mathbb{F}_q(\lambda)$ gives $h \leq h_0$; for $q = 2$, $\kappa = \lambda^{-1}$ gives equality. Thus this family does not hide a new small-first-layer exclusion. For odd q, a nonzero trace-zero $\lambda \in \mathbb{F}_{q^2}$ has $h_0 = 2$ but $\kappa = -1$ and $h = 1$, so the distinction is necessary. □

Example J.6 (Surjectivity is weaker than balance)**.** Lean: `ratioDecoder_surjective`, `ratio_nonzero_fiber_card`, `ratio_zero_fiber_card`, `ratio_eight_fiber_cards` [`CompositionRatio.lean`]
Lean: `ratio_multiplier_conjugate_covariance` [`CompositionMultiplierClasses.lean`]

For $E = \mathbb{F}_8$, $d = 3$, $b = 1$ and any $\lambda \neq 0$, use the preceding actual conjugates. On the complete row-word input E^3, the total map

$$f(x_0, x_1, x_2) = \begin{cases} x_1/x_0 & x_0 \neq 0, \\ 0 & x_0 = 0 \end{cases}$$

is surjective and satisfies $f(Q_c X) = \Phi(f(X))$ for every receiver. Its seven nonzero fibres have size 56 and its zero fibre has size 120, rather than the required common size 64. More generally, over a field of size Q, the corresponding fibre sizes are $(Q-1)Q$ and $Q^2 + (Q-1)Q$.

Outline of the formal proof. Each Q_c applies Frobenius and the same nonzero multiplier to all three rows, so the common multiplier cancels in the quotient; the zero branch is preserved. The input $(1, z, 0)$ attains any z. For nonzero z there are seven choices of x_0 and eight of x_2, with $x_1 = zx_0$ fixed. For $z = 0$, add the 64 inputs with $x_0 = 0$ to the 56 with $x_0 \neq 0, x_1 = 0$. These sum to 512. No complete bijective transposer or balanced decoder is constructed by this example. □

Proposition J.7 (Exact orbit-assignment criterion)**.** Lean: `OrbitAssignment.orbit_map_exists_iff`, `OrbitAssignment.allowed_iff_divides`, `OrbitAssignment.cyclic_balanced_map_iff_orbit_assignment` [`CompositionOrbitAssignment.lean`]

Let a finite group G act on a finite source set and let $\theta : G \twoheadrightarrow \mathbb{Z}/d$ specify the cyclic action on a finite target set. For each source orbit O, choose $x \in O$ and put $n_O = |O|$ and $t_O = [\mathbb{Z}/d : \theta(G_x)]$. For each individual target orbit D, put $\ell_D = |D|$. An equivariant map with every target fibre of size F exists if and only if the whole source orbits can be assigned to individual target orbits, with $\ell_D \mid t_O$ for every assignment and

$$\sum_{O \text{ assigned to } D} \frac{n_O}{\ell_D} = F \quad \textit{for each } D.$$

Outline of the formal proof. A map from the transitive orbit O to D is determined by the image z of x. It is well defined precisely when $\theta(G_x) \subseteq (\mathbb{Z}/d)_z$, equivalently $\ell_D \mid t_O$ by the subgroup structure of a finite cyclic group. When it exists, the map $gx \mapsto \theta(g)z$ is onto D and every fibre has size n_O/ℓ_D: group translation bijects any two fibres, and counting their disjoint union gives the quotient. The phase stabilizer is independent of the source representative because the phase group is abelian. Restricting a balanced map to each source orbit proves necessity. Conversely choose one such orbit map for every assignment; their disjoint union has the stipulated fibre sizes. Whole source orbits cannot be split fractionally. This is an exact test for one class function, not for joint bijectivity of all output columns. □

K Nonlinear corrections to a complete affine baseline

An operator bound becomes useful for a computation only after its *complete* baseline is exhibited. This appendix gives that conditional interface. It allows arbitrary nonlinear corrections and arbitrary correctness sets, but supplies no cheap affine approximation to an arbitrary machine. Fixed-advice fibre counting is already explicit in [3, Theorem 5.5] and [1, Corollary 6.16]. We use the same counting principle with a chronological computation and retain the actual propagated correction span. The distinction prevents charging one correction again at each consumer.

K.1 The complete baseline and its joint record

Let F be a finite field of size q, and let $X \in F^N$ be the raw input. All current payload, retained control and earlier output coordinates belong to the state. A finite sequence of state spaces can be embedded in their direct sum, so we use one finite-dimensional state space V. The baseline initialization, updates and readout are global affine maps

$$v_0 = JX + j, \qquad G_i(v) = L_i v + g_i, \qquad \text{out} = Ov_m + o.$$

Let the complete baseline output be $K_0X + c_0$. At a marked update the actual map may be arbitrary. Its residual on the actual old state is $E_i u_i(X)$, where E_i is a fixed linear insertion map. At other updates the residual is zero. The initialization must also agree with the baseline, or its departure must be included as a marked initial residual. All unmarked maps, including final readout, remain the same *global* affine maps on every reached state.

Choose one preimage $u_i(X)$ in a fixed linear section of $\operatorname{im} E_i$. These values need not be independent or linear in X. Write $Z(X)$ for their one joint tuple, and put

$$A = \tau - K_0, \qquad D = \dim \ker A, \qquad W = \sum_i \operatorname{im}(OL_{m-1} \cdots L_{i+1}E_i),$$

where the empty product is the identity. These are comparison records and subspaces; the definition does not give a native machine free access to them.

Theorem K.1 (Joint residual replay)**.** Lean: `residual_readout_sum`, `actual_readout_sum`, `actual_readout_insertions` [`ResidualPropagation.lean`]

The complete-program counting endpoint is `actual_correct_card_le` *in* ***`ResidualProgram.lean`****. For every fixed tuple $z = (u_i)$, replay gives*

$$\mathrm{out}_z(X) = K_0X + c_0 + \sum_i OL_{m-1}\cdots L_{i+1}E_iu_i.$$

On an arbitrary correctness set C, the actual tuple $Z(X)$ reproduces the actual computation. Consequently

$$|C| \le |Z(C)|q^D. \tag{33}$$

Outline of the formal proof. Induct in the original update order. After fixing the residual values, each update is $v \mapsto L_iv + g_i + E_iu_i$, so its linear part is still L_i. Expanding the recurrence propagates each inserted residual through exactly the subsequent linear maps and then through O. With $z = Z(X)$ the recurrence agrees with every actual update on X, including retained outputs. On a fixed record fibre, correctness is the affine equation $AX = c(z)$. Subtract one solution, if there is one, to inject the fibre into $\ker A$. It therefore contains at most q^D elements. Sum over the *attained* joint records. The formal proofs perform this recurrence and translate-to-kernel injection; they do not assume that C is a subspace. □

Theorem K.2 (Joint output-span bound)**.** Lean: `residualSpanDescriptor_injective`, `finrank_preimage_le_kernel_add_intersection`, `residual_span_card_le` [`ResidualSpan.lean`]
Lean: `actual_correct_span_card_le` [`ResidualOutputSpan.lean`]

For the actual program, and more generally, if $C \subseteq F^N$ satisfies $AX - c \in W$ for every $X \in C$, then

$$|C| \le q^{D+\dim(W\cap \mathrm{im}\, A)} \le q^{D+\dim W}. \tag{34}$$

For the replay above, take $c = c_0$ and the displayed propagated span W.

Outline of the formal proof. Subtract one point of C to inject it into $A^{-1}(W)$. Choose a linear projection onto $\ker A$ that is the identity on that kernel. The map sending $x \in A^{-1}(W)$ to its projection and Ax is injective, with values in $\ker A \times (W \cap \mathrm{im}\, A)$. This proves the dimension bound and then the finite-field cardinality bound. In particular, the invoice uses the joint propagated span, not a sum that charges dependent copies repeatedly. Its dimension is at most the sum of the insertion ranks. □

Corollary K.3 (Information and sparse corrections)**.** Lean: `residual_entropy_lower`, `residual_uniform_entropy_lower`, `actual_correct_entropy_lower`, `residual_span_information` [`ResidualEntropy.lean`]

The sparse-count endpoint is `actual_sparse_card_le` *in* ***`ResidualSparse.lean`****. For any law supported on C, with entropy in base q,*

$$H_q(Z) \ge H_q(X) - D.$$

If C is uniform and $\Delta = N - \log_q|C|$, this gives $H_q(Z) \ge N - D - \Delta$ and $\dim W \ge N - D - \Delta$. If each of m public insertion ports has rank at most r, the correctly handled inputs on which at most s residuals are nonzero number at most

$$q^D \sum_{j=0}^{\min(s,m)} \binom{m}{j}(q^r - 1)^j. \tag{35}$$

Outline of the formal proof. Each conditional input fibre has at most q^D points, so $H_q(X \mid Z) \le D$. Since Z is a function of X, the entropy chain rule gives the first inequality. The span bound gives the second assertion for uniform C. For the sparse count, choose coordinates in each insertion image, with the zero residual represented uniquely by zero. A support of j positions allows at most $(q^r - 1)^j$ nonzero value tuples. Sum over supports and apply Eq. (33). The locations may depend arbitrarily on X; their support count is already included. The formal count records each actual residual as a value in its insertion image, so it charges the insertion rank and does not assume a particular preimage selection. The shared support/value encoding also permits different alphabets at different ports. □

For fixed q, r, $m = O(N)$ and $s = o(N)$, the logarithm of the sum in Eq. (35) is $o(N)$. The formal endpoint is `LeanMachine.sparseMenu_log_div_tendsto_zero`. Its finite proof multiplies the sum by x^s and bounds the result by $(1 + Ax)^m$ for $0 < x \le 1$, where $A = q^r - 1$. Choose x small enough that $m \log(1 + Ax)$ costs at most half the desired linear budget, then bound $s \log(1/x)$ by the other half. This treats $s = 0$ directly and requires no lower bound on m/N.

K.2 What can be substituted, and what remains missing

Suppose $N = bd^2$ and the *complete* baseline has its column factors in one common triangularizable algebra. The flag bound and its entry-width extension give $D \le bd(d+1)/2$[1, corollary "One triangularizable side" and remark "Entries of b bits"]. Thus correct inputs of deficit Δ require

$$H_q(Z) \ge \frac{bd(d-1)}{2} - \Delta, \qquad \dim W \ge \frac{bd(d-1)}{2} - \Delta.$$

An independently established sharper operator bound may replace this D. With fixed-width insertion events, sublinear nonzero corrections therefore cover at most $q^{(1/2+o(1))N}$ inputs. A whole-row correction of rank $O(db)$ is not a fixed-width event.

The revised entry-rule bound of [1, Proposition 6.15 and Corollary 6.16] is another available specialization when its actual entry functionals are supplied. It prices advice by its support and changes of an entry decoder by half a symbol. A change of a whole row is d entry changes. Additive advice alone need not change any decoder. Our rank invoice and that change count are different statistics; neither may be substituted for the other without checking the complete rule.

An input-dependent branch or schema can be included in the joint record. Then each branch has its own complete baseline and kernel bound D_θ; the support estimate sums the corresponding q^{D_θ} fibre bounds, and entropy gives $H_q(Z) \ge H_q(X) - \mathbb{E}D_\theta$. These attained-branch and entropy bounds are `residual_branch_card_le` and `residual_branch_entropy_lower` in `ResidualBranches.lean`. Unknown earlier encodings, address histories, mutable flags or nonlinear unmarked readout must be included in such a proved interface. They cannot be omitted by checking only the zero-residual trajectory.

These statements exclude sparse corrections to an *exhibited* structured computation. They do not show that every fast transposer is close to one, or that producing a correction of large rank costs superlinear time. Supplying that structural comparison with a small joint record remains the construction/impossibility bridge.

K.3 Two sharp complete-baseline examples

Proposition K.4 (Separable baseline defect)**.** Lean: `Separable.rank_defect_lower` [`ResidualSeparable.lean`]
Lean: `Separable.initialProjection_rank`, `Separable.rank_defect_sharp` [`ResidualSeparableSharp.lean`]

For $L(X) = AXB^{\mathsf{T}}$ over any field, put $r = \min(\operatorname{rank} A, \operatorname{rank} B)$. Then

$$\operatorname{rank}(\tau - L) \geq d^2 - \frac{r(r+1)}{2}.$$

Equality is attained for $A = B = \operatorname{diag}(I_r, 0)$ in every characteristic. Thus this is a sharp bound for one separable baseline, not for a sum of arbitrary tensor terms.

Outline of the formal proof. Writing $Z = X^{\mathsf{T}}$ turns agreement into $Z = AZ^{\mathsf{T}}B^{\mathsf{T}}$. If $a = \operatorname{rank} A$, choose invertible U, V with $UAV^{-1} = E_a$. For $Y = UZV^{\mathsf{T}}$ and $D = VBU^{-1}$ the equation is $Y = E_a Y^{\mathsf{T}} D^{\mathsf{T}}$. The last $d - a$ rows vanish, and the upper-left a-square block Q determines the others and satisfies $Q = Q^{\mathsf{T}} D_{11}^{\mathsf{T}}$. Its lower triangle, including the diagonal, determines Q: it gives column zero and therefore row zero; the recovered rows and the lower triangle then give the next column and row, successively. The formal proof obtains the invertible factors and the active coordinate reindexing from the actual matrix rank normal form. The resulting restriction is a linear injection into the unordered-pair coordinates of the active block, so the kernel has dimension at most $a(a+1)/2$. Transposing the agreement equation exchanges A, B and gives the same bound with $a = \operatorname{rank} B$. Rank–nullity proves the claim. For $A = B = E_r$, agreement is exactly symmetry inside the leading r-square block and zero outside, with $r(r+1)/2$ independent entries. □

Proposition K.5 (Nonlinear row-then-column agreement)**.** Lean: `RowColumn.agreement_card_le`, `RowColumn.helper_agreement_card_le`, `RowColumn.identity_agreement_card` [`ResidualRowColumn.lean`]
Lean: `RowColumn.exists_sharp_helper`, `RowColumn.helper_bound_attained` [`ResidualRowColumnSharp.lean`]

Over an alphabet of size $q \geq 2$, first apply an arbitrary fixed map to each row, then an arbitrary fixed map to each column. The resulting complete map agrees with transpose on at most $q^{d(d+1)/2}$ inputs. If one joint helper chooses all these maps, at most $|\operatorname{range} H| q^{d(d+1)/2}$ inputs can be correct. Both bounds are sharp as abstract counting statements.

Outline of the formal proof. Write the intermediate matrix as $Y_{j,:} = R_j(X_{j,:})$ and the output column as $C_j(Y_{:,j})$. On agreement inputs, $X_{j,:} = C_j(Y_{:,j})$ and therefore $Y_{j,:} = R_j(C_j(Y_{:,j}))$. The lower triangle of Y determines column zero, hence row zero; then column one, hence row one; continue in order. It determines all of Y, which in turn determines X. This injection proves the first count without assuming either family is linear or injective. Fix the helper and sum to prove the second count. Identity row/column maps attain the first bound on symmetric inputs. For the second, identify the alphabet with the cyclic group of order q and let the helper give $X_{ij} - X_{ji}$ for $i < j$. Identity row maps followed by column maps adding or subtracting the corresponding differences transpose every input. Every helper tuple is attained and there are $q^{d(d-1)/2}$ tuples. Computing and delivering this helper is not free in a native model. □

L Formalization boundary

The library uses the earlier shared-decoder models and Mathlib, with the repository's existing pins. Its entry type is a module, so vector lanes may be mixed unless a theorem explicitly imposes

additional structure. Arbitrary deterministic shared rules and linear shared rules are separate predicates. Complete factorization theorems quantify over total maps on the full state space. Bijectivity follows from exactness and finite dimension for linear factors, and from exactness and a finite state space for nonlinear factors.

The finite capacity proof proceeds through actual translated source spaces, their component quotient, and cyclic run counts. Its arithmetic averaging lemma is separate from those structural statements. The composition proof similarly derives its forced row-class subspace from an actual factorization; the common-message inequality is not an assumed surrogate for that step. Finite progression extraction reuses Mathlib's Hales–Jewett theorem. The small two-color certificate is evaluated by the kernel, without native evaluation. Some finite checks use a larger heartbeat allowance.

All numbered mathematical statements, except the native implementation theorem, are now checked. This includes exact arbitrary-set capacities and their cyclic quotient counts, line-marker constructions, every structured and probabilistic mask estimate, the actual scalar-extension/Fourier bridge for the reflection barrier, multiplier and whole-orbit diagnostics, and sharp separable and nonlinear row-then-column examples. The reflected linear-subcode bound holds at every odd dimension; primality is retained only where the separate construction needs it.

The finite probability library constructs the actual uniform source law, proves its bounded-differences inequalities and computes the required event expectations and coordinate sensitivities. Conditioning, unlabelled partitions and simultaneous run-spectrum events are included. The residual library evaluates the complete chronological program, propagates its literal deviations, and proves joint record, entropy, span and input-dependent sparse-support bounds. These statements retain correlation between records and do not assume that a correctness source is a subspace. The generated index lists the exact endpoints. Finite seed coefficients are checked by the Lean kernel; the external generator is not trusted.

The native implementation has a distinct large obligation: implementing the two-tape protocol as a transition table in the frozen model, with a complete execution invariant and uniform time and state accounting. Its mathematical protocol is proved in the text, but is not certified by an algebraic code or a finite dimension calculation. That compiler is recorded as a substantial formalization todo in the README. No additional axiom stands in for it.

M Index of Lean declarations

The following declarations are cited by formal-status lines. Names are relative to `SharedDecoders`. A partial status retains the gap stated at its theorem.

`CapacityAmbient.lean` `IsSharedCode.component_finrank_le`, `covarianceSpace_rot`, `finrank_covarianceSpace`.

`CapacityClosureAsymptotic.lean` `CapacityClosure.capacity_bound`, `CapacityClosure.error_ratio_tendsto`, `CapacityClosure.residual_finite_bound`, `CapacityClosure.rounds_eq_floor`, `CapacityClosure.uniform_eventual_bound`.

`CapacityClosureControl.lean` `CapacityClosure.scalarHull_supportedSymmetric`, `CapacityClosure.symmetric_copy_defect`, `CapacityClosure.symmetric_isSharedCode`.

`CapacityCompletion.lean` `IsSharedCode.completedCode`, `completedCode_eq_of_range_le`, `finrank_completedCode_add_rank`, `finrank_le_completedCode`.

CapacityControls.lean CapacityControls.exists_deterministic_not_linear.

CapacityDifferenceCover.lean IsSharedCode.difference_cover_real_bound, IsSharedCode.saturated_cover_bound, IsSharedCode.square_cover_bound.

CapacityDigitParityEnvelope.lean CapacityDigitParity.digit_parity_envelope, CapacityDigitParity.digit_parity_floor_envelope.

CapacityFieldLinearComponents.lean CapacityFieldLinear.component_of_scalarTest, CapacityFieldLinear.component_zero_of_scalarTest_zero, CapacityFieldLinear.components_unique, CapacityFieldLinear.phaseBasis, CapacityFieldLinear.sum_components.

CapacityFieldLinearCorollaries.lean CapacityFieldLinear.one_class_original_bound, CapacityFieldLinear.one_class_residual_eq_bot, CapacityFieldLinear.two_class_half_bound, CapacityFieldLinear.two_class_original_bound.

CapacityFieldLinearResidual.lean CapacityFieldLinear.active_class_residual_bound, CapacityFieldLinear.finrank_encodedResidual, CapacityFieldLinear.original_source_class_bound.

CapacityIntervals.lean CapacityIntervals.exists_binary_code_dimension, CapacityIntervals.exists_optimal_interval_code, IsSharedCode.two_mul_finrank_le_interval.

CapacityIteration.lean IsSharedCode.iterated_finrank_le, componentCount_zero.

CapacityLineDeterministic.lean IsDeterministicLineCode.affine_two_mul_finrank_le, IsDeterministicLineCode.two_mul_finrank_le.

CapacityLineMarker.lean LineMarker.affineMarker_finrank, LineMarker.code_card, LineMarker.exists_decoder.

CapacityLineMarkerRate.lean LineMarker.code_density, LineMarker.code_density_tendsto.

CapacityLinearization.lean IsLinearSharedCode.inr_comap, linear_subcode_lane_bound.

CapacityMarker.lean CapacityMarker.exists_affine_source_decoder.

CapacityMaskComparisonProbability.lean
CapacityMaskProbability.ChangeGood.certificate_floor, CapacityMaskProbability.EqualityGood.direct_gain_inverse_sqrt, CapacityMaskProbability.component_floor_probability, CapacityMaskProbability.direct_floor_probability.

CapacityMaskFinite.lean IsSharedCode.four_mul_finrank_le, IsSharedCode.six_mul_finrank_le, finite_weighted_run_budget.

CapacityMaskMethodEndpoints.lean CapacityMaskProbability.IsolationGood.rounded_floor, CapacityMaskProbability.isolation_rate_probability.

CapacityMaskRunProbability.lean CapacityMaskProbability.SpectrumGood.source_bound, CapacityMaskProbability.spectrum_bad_probability.

CapacityMaskRunSpectrum.lean IsSharedCode.run_spectrum_three.

CapacityMaskStructured.lean CapacityMaskStructured.first_component_count, CapacityMaskStructured.second_component_count, CapacityMaskStructured.structured_finrank_le.

CapacityOptimization.lean capacity_transversal_deficiency, linear_capacity_eq_residual, optimal_transversal_full_rank, shared_capacity_eq_residual.

CapacityReflectionBarrier.lean CapacityReflection.deletion_rate_limit, CapacityReflection.one_class_finrank_le, CapacityReflection.two_class_codimension_lower, CapacityReflection.two_class_finrank_le, CapacityReflection.two_class_positive_deletion.

CapacityReflectionSource.lean CapacityReflection.finrank_rowSpace, CapacityReflection.finrank_source, CapacityReflection.source_disjoint_views, CapacityReflection.source_shared.

CapacityRetentionAverage.lean IsSharedCode.retention_average, IsSharedCode.retention_normalized.

CapacityRetentionControl.lean RetentionControl.laneSwap_eq_transpose_iff, RetentionControl.paired_source_rate_and_loss.

CapacityRetentionCoprime.lean IsSharedCode.eq_symmetric_of_at_most_one_exception, IsSharedCode.full_symmetric_retention, IsSharedCode.prime_full_symmetric_retention.

CapacityRetentionFinite.lean IsSharedCode.retention_bound.

CapacityRetentionStability.lean RetentionAverage.fixed_gain_requires_loss, RetentionAverage.stability, RetentionAverage.stability_bounded_classes.

CapacityRuns.lean CapacityRuns.changeCount_double, CapacityRuns.weighted_run_budget.

CapacityRunsComponents.lean componentCount_le_changes_add_gcd, componentCount_le_runs_add_two_gcd.

CapacitySetCounting.lean SetCapacity.fixed_H_card, SetCapacity.fixed_K_card, SetCapacity.prime_capacity.

CapacitySetOrbit.lean SetCapacity.capacity_average_formula, SetCapacity.capacity_orbit_formula, SetCapacity.exact_capacity.

CapacitySetRate.lean SetCapacity.capacity_lower, SetCapacity.logarithmic_rate_tendsto, SetCapacity.relative_error.

CapacitySubcodeBarrier.lean CapacitySubcode.above_half_seed_barrier.

CapacitySubcodeMixture.lean CapacitySubcode.exists_above_half_barrier.

CapacityTripleBinary.lean CapacityTriple.exists_width_stable_hard_source.

CapacityUniversal.lean IsSharedCode.two_class_capacity_bound.

CompositionAgreement.lean full_input_agreement_probability, repeated_class_agreement_probability.

CompositionAnchoredEndpoint.lean anchored_adjacent_classes_distinct, anchored_at_most_two_classes_impossible.

CompositionChannels.lean commonMessage_interval_bound.

CompositionEndpointBlowup.lean endpointFirst_classUniform, endpointFirst_involutive, endpointSecond_equivariant, endpointSecond_involutive, endpoint_composite_getter, endpoint_composite_requires_singletons.

CompositionLinear.lean exists_twoHop_cutoff, twoHop_coprime_progression_dimension_le, twoHop_dimension_le, twoHop_progression_gcd, twoHop_two_classes_le_sixteen.

CompositionMultiplierClasses.lean fieldMultiplierConjugate_eq, multiplier_class_count, multiplier_class_count_prefix, multiplier_class_residue, ratio_multiplier_conjugate_covariance.

CompositionMultiplierGetters.lean actual_multiplier_getters_card, multiplier_binary_degree_eq, multiplier_degree_le.

CompositionOrbitAssignment.lean OrbitAssignment.allowed_iff_divides, OrbitAssignment.cyclic_balanced_map_iff_orbit_assignment, OrbitAssignment.orbit_map_exists_iff.

CompositionOrbitResidues.lean balanced_fixed_word_residue, balanced_fixed_word_small_impossible, fixedTupleEquiv, invariant_fiber_fixed_residue, no_balanced_free_off_zero, no_balanced_semiregular, tuple_orbit_divisor.

CompositionOrderFive.lean orderFivePerm_fixed, orderFivePerm_order, orderFivePerm_zero, orderFive_adjacent_class_impossible, orderFive_conjugate_difference, orderFive_no_balanced_covariance, orderFive_pair_nontransitive.

CompositionProgression.lean commonMessage_finrank_le.

CompositionRankField.lean exists_twoHop_rank_threshold, twoHop_rank_interval_any_field, twoHop_rank_superlinear.

CompositionRatio.lean ratioDecoder_surjective, ratio_eight_fiber_cards, ratio_nonzero_fiber_card, ratio_zero_fiber_card.

CompositionRecurrence.lean recurrence_finrank_le.

CompositionRobustAgreement.lean agreement_error_cover, robust_full_input_count, robust_repeated_class_count.

CompositionScatteredCompletion.lean scatteredCompletion_equivariant, scatteredFirstFactor_classUniform, scatteredFirstFactor_fixed_preimage, scatteredFirstFactor_source, scattered_one_class_impossible.

CompositionScatteredSource.lean scatteredSource_finrank, scattered_orbit_injective, scattered_punctured_orbits_disjoint.

CompositionSingleSiteEndpoint.lean single_at_most_two_classes_impossible.

CompositionSparseObstruction.lean scatteredFirstFactor_disagreement, scattered_remaining_factor_singletons.

CompositionSparsePatch.lean columnPatch_involutive, columnPatch_nonzero_column, columnPatch_support_card.

CompositionWordAction.lean transitive_pair_impossible.

ConstructionBlockDefects.lean finrank_upperBlockAgreement_le, upperBlockDescriptor_injective.

ConstructionCapacity.lean periodic_linear_capacity_algebraicClosure_upper, periodic_linear_capacity_lower, periodic_linear_capacity_split.

ConstructionCyclicExact.lean binaryCyclicDefect_ker, binarySeed_full_rule, finrank_binaryCyclicLaneDefect_ker, mem_cyclicLiftCode_iff_full_rule.

ConstructionLinearLift.lean cyclic_lift.

ConstructionSeed.lean binaryCyclicCode_interval_excess, binaryCyclicCode_linear, binaryCyclicPartition_surjective, finrank_binaryCyclicCode.

Model.lean ClassUniform, IsSharedCode.

PeriodTransport.lean fixed_conjugate_iff, fixed_image_preserved, literal_period_complete.

ResidualEntropy.lean actual_correct_entropy_lower, residual_entropy_lower, residual_span_information, residual_uniform_entropy_lower.

ResidualOutputSpan.lean actual_correct_span_card_le.

ResidualPropagation.lean actual_readout_insertions, actual_readout_sum, residual_readout_sum.

ResidualRowColumn.lean RowColumn.agreement_card_le, RowColumn.helper_agreement_card_le, RowColumn.identity_agreement_card.

ResidualRowColumnSharp.lean RowColumn.exists_sharp_helper, RowColumn.helper_bound_attained.

ResidualSeparable.lean Separable.rank_defect_lower.

ResidualSeparableSharp.lean Separable.initialProjection_rank, Separable.rank_defect_sharp.

ResidualSpan.lean finrank_preimage_le_kernel_add_intersection, residualSpanDescriptor_injective, residual_span_card_le.

SubgroupClasses.lean covariance_no_class_collision, exact_periodic_first_class_bound, exact_periodic_first_class_size_bound, orbit_classes_injective.

UniformSandwich.lean uniform_sandwich_finite, uniform_sandwich_injective_classes.

Other declarations named in the text CapacityTernarySpectrum.Good.source_bound, CapacityTernarySpectrum.bad_probability, IsSharedCode.intrinsic_retention_bound, LineMarker.affineMarker_exceeds_half, actual_correct_card_le, actual_sparse_card_le, binary_progression_nine_finite, exists_progression_bound, mem_liftedCode_iff_full_rule, residual_branch_card_le, residual_branch_entropy_lower.

Imported declarations LeanMachine.sparseMenu_log_div_tendsto_zero.

N Building and verifying

The repository must be cloned with its sibling paper directories. The Lean package pins Lean and Mathlib to version 4.34.1 and imports earlier libraries through relative paths. From the paper directory, run

```
python3 tools/lean_index.py
cd lean
lake exe cache get
lake build
lake env lean PaperAxioms.lean
cd ../paper
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
```

The generated index and axiom file list every declaration cited by a formal-status line. The permitted axioms are `propext`, `Classical.choice` and `Quot.sound`. There are no admitted proofs, extra axioms or native computation shortcuts. Build products under `.lake/` are local and are not part of the published sources. The README records the remaining formalization obligations; a successful build does not certify the written-only parts.

One quotient-probability proof uses a local `maxHeartbeats` limit of 800000 for elaboration. This does not change its statement or kernel checking; the remaining probability modules use the default limit.

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